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Section: 11

PRACTICAL – 2

Creating a Virtual Machine, Instances and an S3 Bucket on AWS

- **Part 1 – Virtual Machine:** launching a Linux virtual machine with Amazon EC2 and working on it from the terminal.
- **Part 2 – Instances:** launching a Windows Server instance (connected by Remote Desktop) and an Ubuntu instance with a graphical desktop (connected by Remote Desktop).
- **Part 3 – S3 Bucket:** creating a storage space in Amazon S3.

PART 1: CREATING A LINUX VIRTUAL MACHINE USING AWS EC2

EC2 (Elastic Compute Cloud) is an Amazon Web Services (AWS) service that lets users create and manage virtual machines (virtual servers) in the cloud. A **Virtual Machine (VM)** is a software-based computer that provides its own CPU, RAM (memory), storage, network connectivity and operating system. In this part a Linux virtual machine is created with Amazon EC2.

Step 1: Open the EC2 console

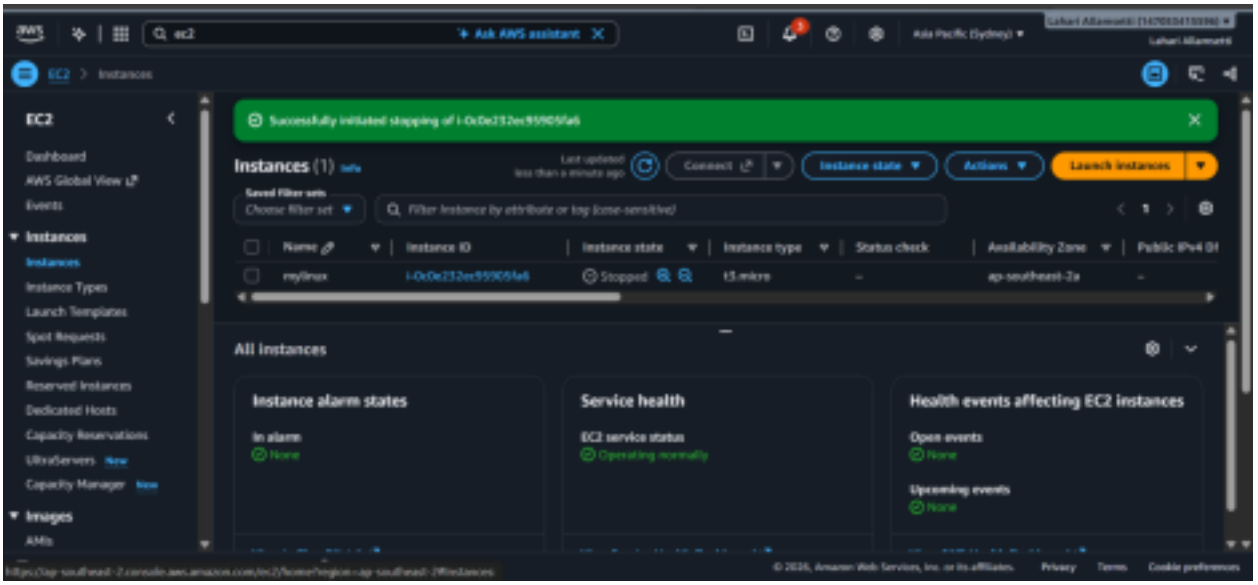
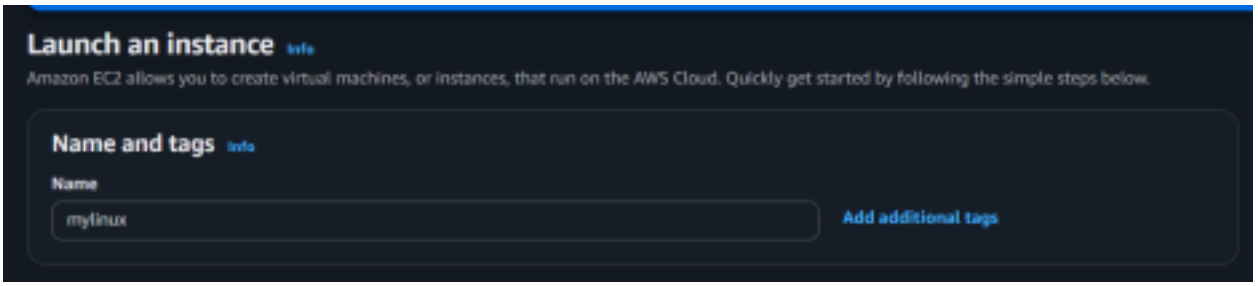


Fig: EC2 Instances page

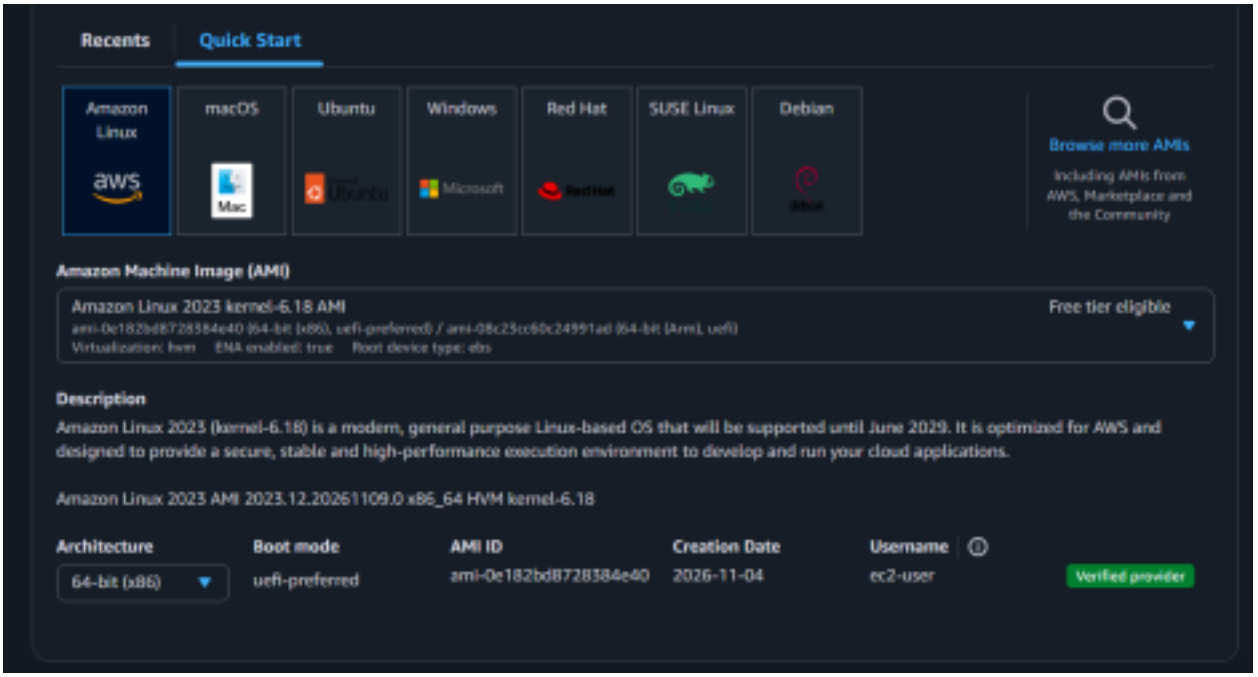
After signing in, **ec2** is typed in the search bar and the EC2 service is opened. The **Instances** page lists every virtual machine in the account along with its ID, state, type and Availability Zone. The orange **Launch instances** button at the top right starts the creation of a new virtual machine. The green banner is only a confirmation of an earlier stop action on an existing instance.

Step 2: Name the instance



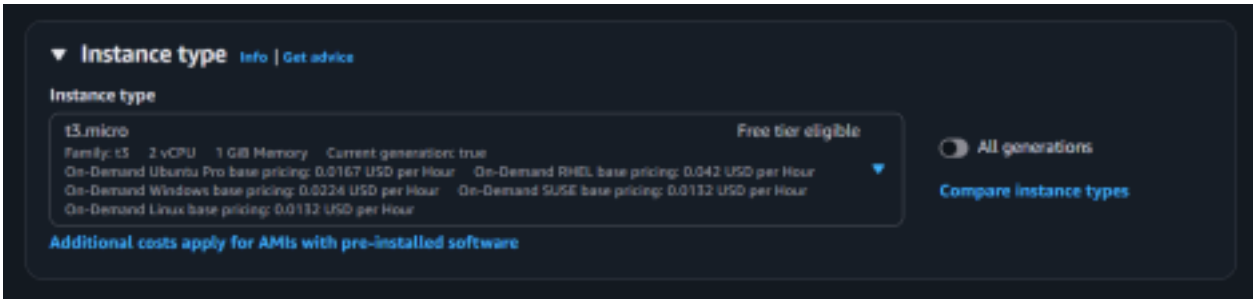
The **Launch an instance** wizard opens. In **Name and tags** the instance is given the name **mylinux**, so that it can be recognised easily in the instance list. Extra tags can be added with **Add additional tags**.

Step 3: Select an Amazon Machine Image (AMI)



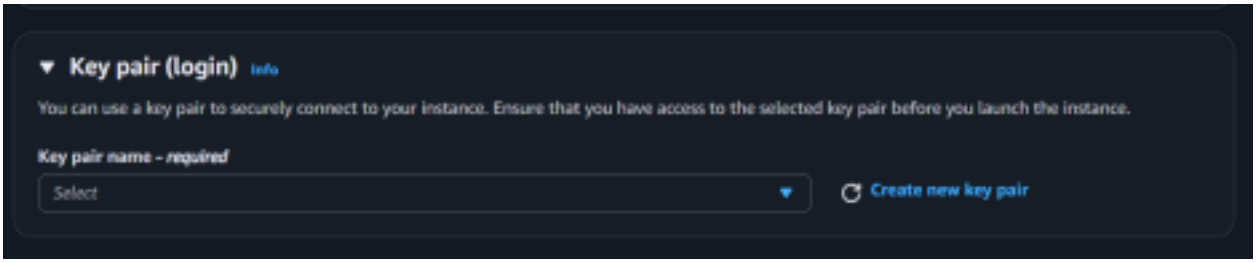
An **AMI (Amazon Machine Image)** is a template that contains the operating system and the configuration needed to launch a virtual machine.

Step 4: Choose the instance type



The instance type decides the CPU and memory given to the virtual machine. **t2.micro** and **t3.micro** are small, burstable, general-purpose types. Here **t3.micro** is selected, which is enough for running applications and basic Linux operations.

Step 5: Key pair (login)



A key pair is needed to connect to the instance securely. As no key pair exists yet, **Create new key pair** is clicked from this section. The instance cannot be launched until a key pair has been selected.

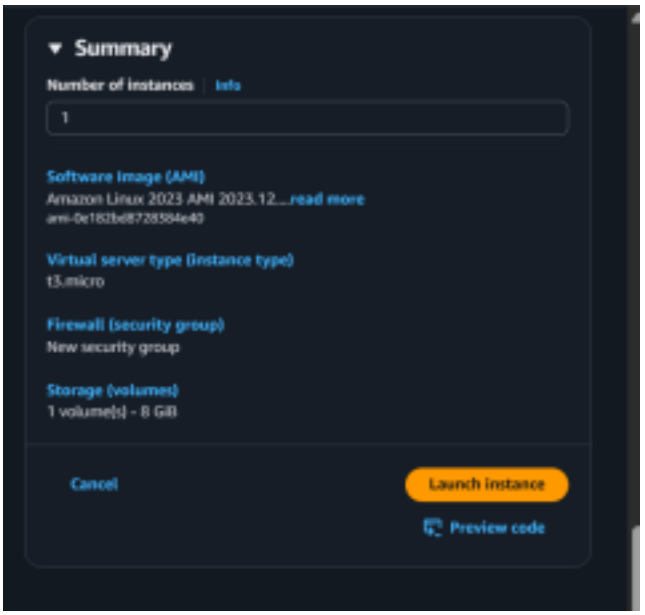
Create the key pair



A **key pair** has a **public key** and a **private key** . To create it: (1) enter a name (**key1**), (2) select the **RSA** type, (3) choose the **.pem** private key format (used with OpenSSH; .ppk is for PuTTY), (4) click **Create key pair**, and (5) save the downloaded file securely, because it is needed later to connect.

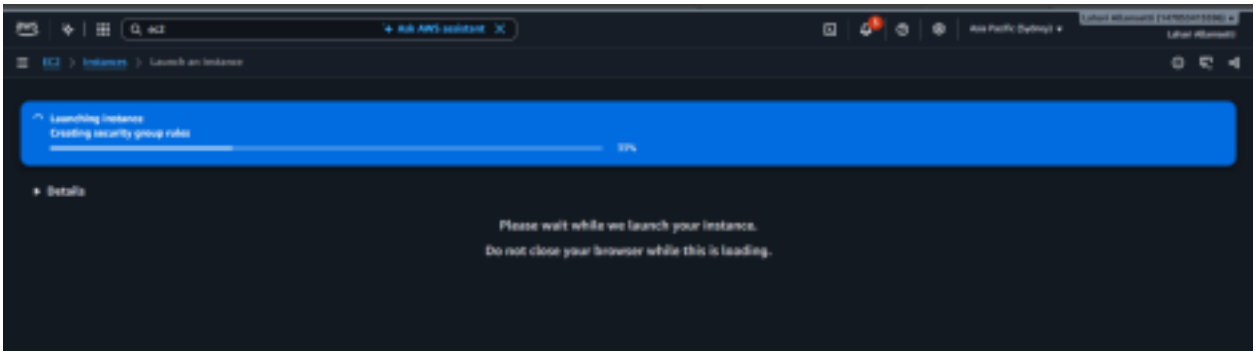
RSA and SSH in brief. RSA is a public-key algorithm that uses a public key and a private key for secure communication. Encryption converts readable data into a coded form to protect it, and decryption converts it back. **SSH (Secure Shell)** is the network protocol (port 22) used to log in to and manage a remote server securely, with encrypted communication and key-pair authentication. SSH only connects to a server that already exists; the EC2 instance itself is created with the Launch Instance option.

Step 6: Review the summary and launch



The **Summary** panel shows the final configuration: number of instances **1**, Amazon Linux 2023 AMI, instance type **t3.micro**, a **new security group** (the virtual firewall) and one **8 GiB** storage volume. After checking these details, **Launch instance** is clicked on the right.

Launch in progress



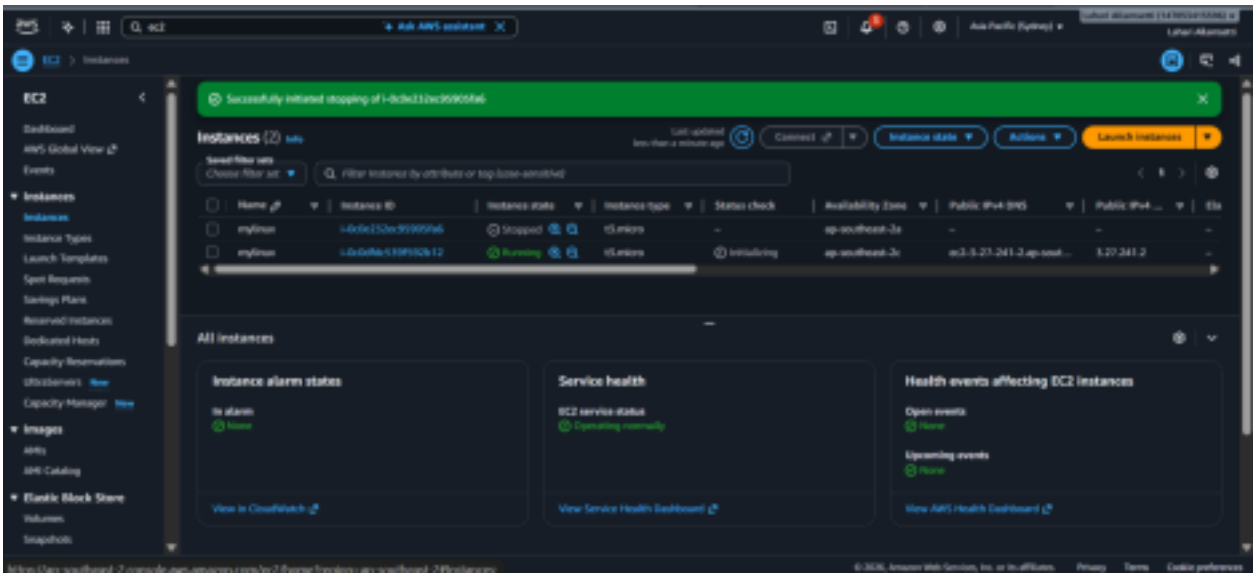
AWS now creates the virtual machine. The progress bar shows the tasks it is doing, such as Creating security group rules. The browser must not be closed until this finishes.

Launch successful



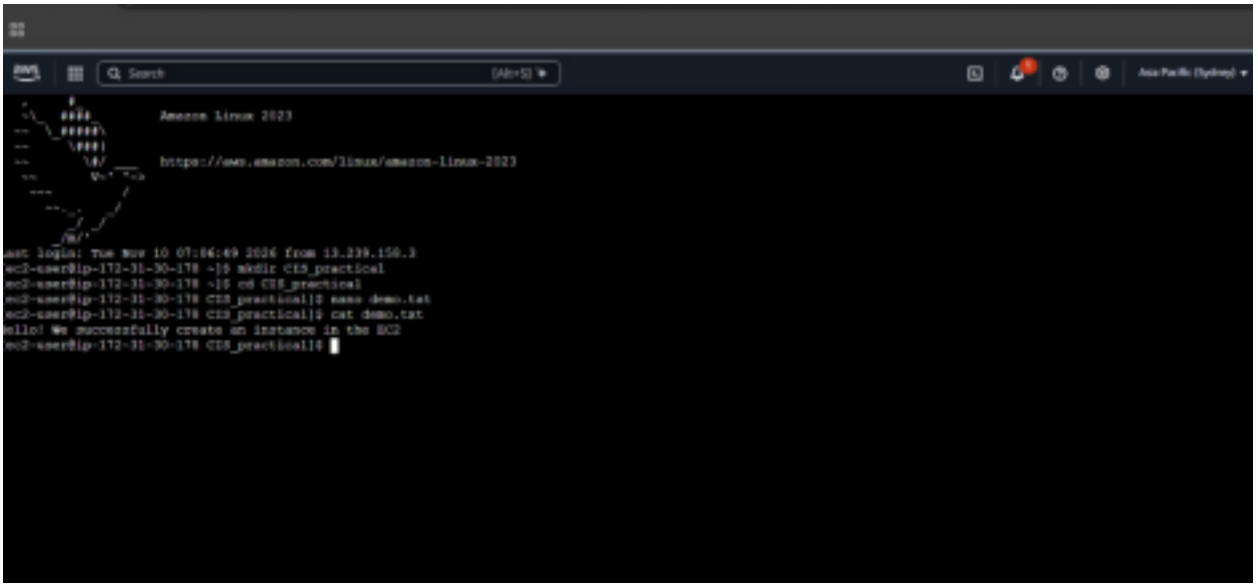
The green **Success** banner says the launch of the instance was initiated, together with the new instance ID. The virtual machine now exists and is starting up.

Step 7: Check that the instance is running



On the **Instances** page the new **mylinux** instance shows the state **Running** with the status check Initializing. AWS runs **status checks** to confirm the instance and its supporting resources work properly; the instance is ready when the checks show as passed (for example 3/3 or 2/2). To connect, the instance checkbox is selected and **Connect** is clicked, and then a method such as **EC2 Instance Connect** or an SSH client is chosen.

Step 8: Basic Linux operations on the instance



The terminal opens with the **Amazon Linux 2023** welcome banner, so the connection to the remote server is successful. The commands used were: **mkdir CIS_practical** (create a directory), **cd CIS_practical** (move into it), **nano demo.txt** (create and write a file) and **cat demo.txt** (display its content). The file displayed the message that an instance was successfully created in EC2.

Linux commands used

Command	Purpose	Example
mkdir	Creates a new directory	mkdir CIS_practical
cd	Moves to another directory	cd CIS_practical

touch	Creates an empty file	touch notes.txt
echo ... > file	Writes text into a file (>> appends instead of overwriting)	echo "Welcome to Linux EC2" > notes.txt
cat	Displays the content of a file	cat notes.txt
ls	Lists files and directories	ls
pwd	Shows the present working directory	pwd

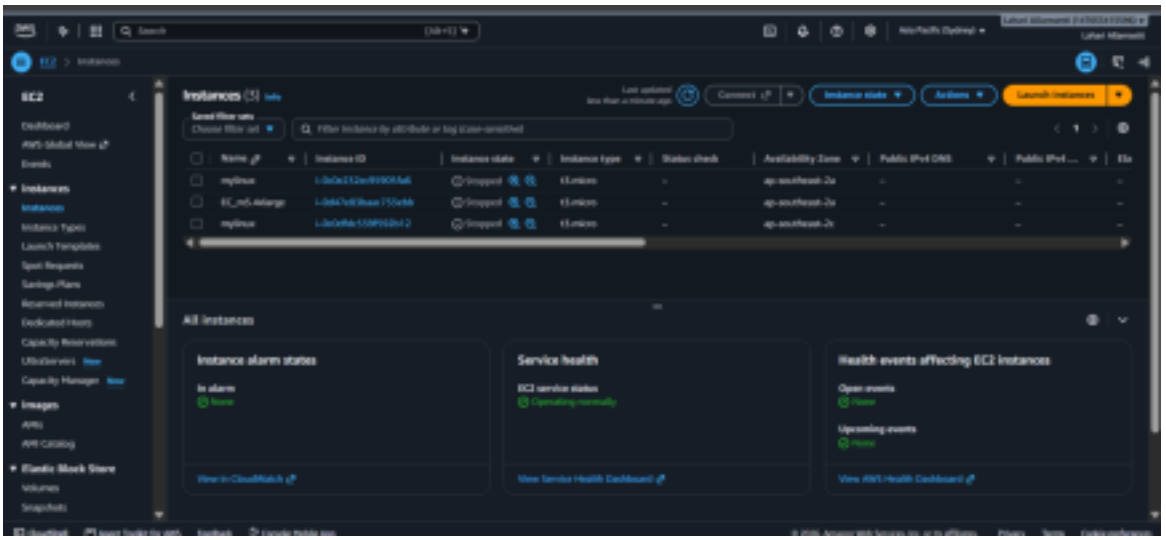
Result: A Linux virtual machine was created and launched on Amazon EC2 using Amazon Linux 2023 and the t3.micro instance type. An RSA key pair was created for secure authentication, the instance was connected to, and basic Linux operations (creating directories and files, writing content and displaying it) were carried out.

PART 2: CREATING INSTANCES (WINDOWS AND LINUX WITH GUI)

In this part two more EC2 instances are launched. A **Windows Server** instance is reached through **Remote Desktop (RDP)**, and an **Ubuntu** instance is set up with a graphical desktop that is also reached through Remote Desktop. **RDP** is the protocol, using port **3389**, that shows the desktop of a remote computer on the local screen.

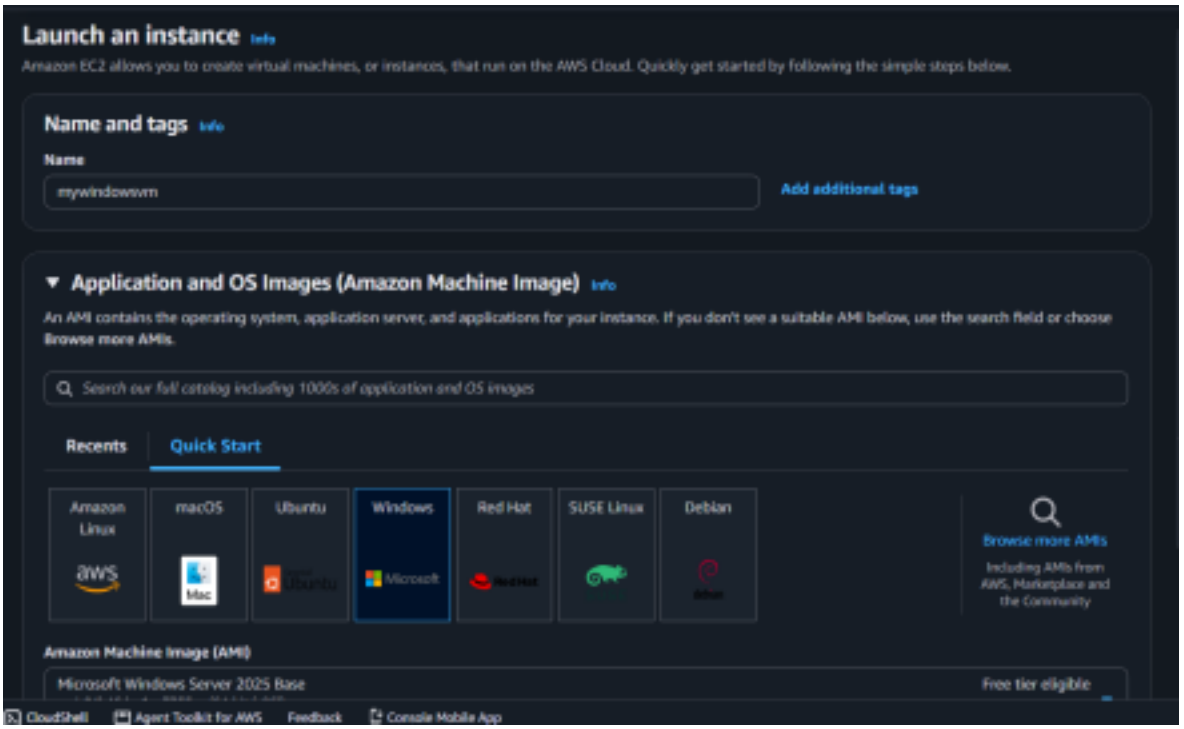
2A. Windows Server instance

Step 1: Open the Instances page



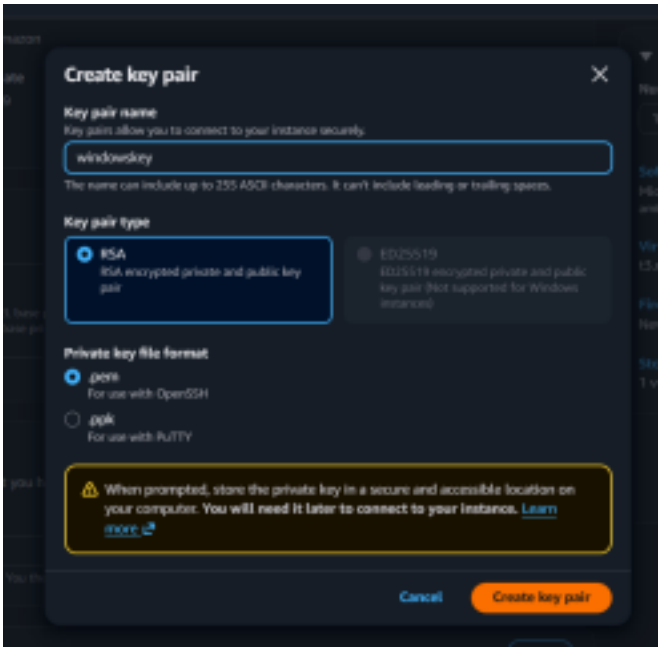
The EC2 **Instances** page lists the instances created earlier (all currently stopped). A new instance is started with **Launch instances**.

Step 2: Name the instance and select the Windows OS



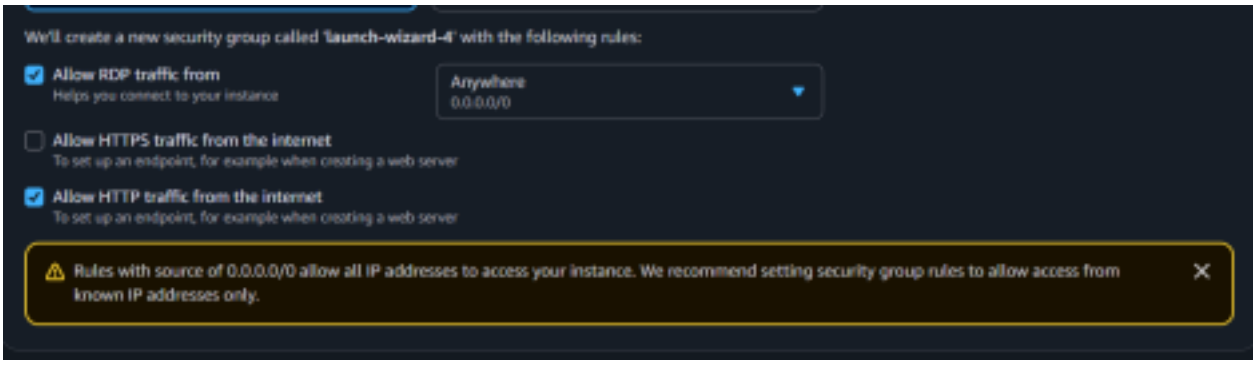
The instance is named **mywindowsvm**. In the Quick Start tabs the **Windows** tile is selected, which gives the AMI **Microsoft Windows Server 2025 Base** (free tier eligible).

Step 3: Create a key pair for Windows



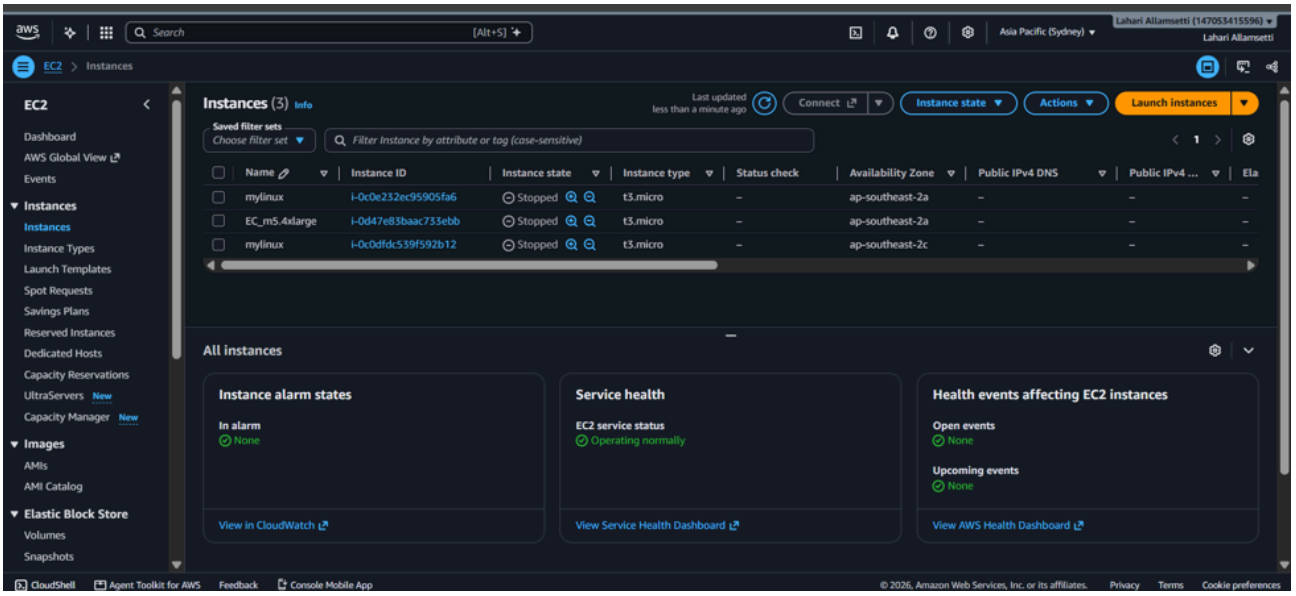
A new key pair named **windowskey** is created with the **RSA** type and the **.pem** format. The ED25519 type is greyed out because it is not supported for Windows instances. The private key file must be saved, because Windows uses it to decrypt the Administrator password.

Step 4: Network settings (security group)



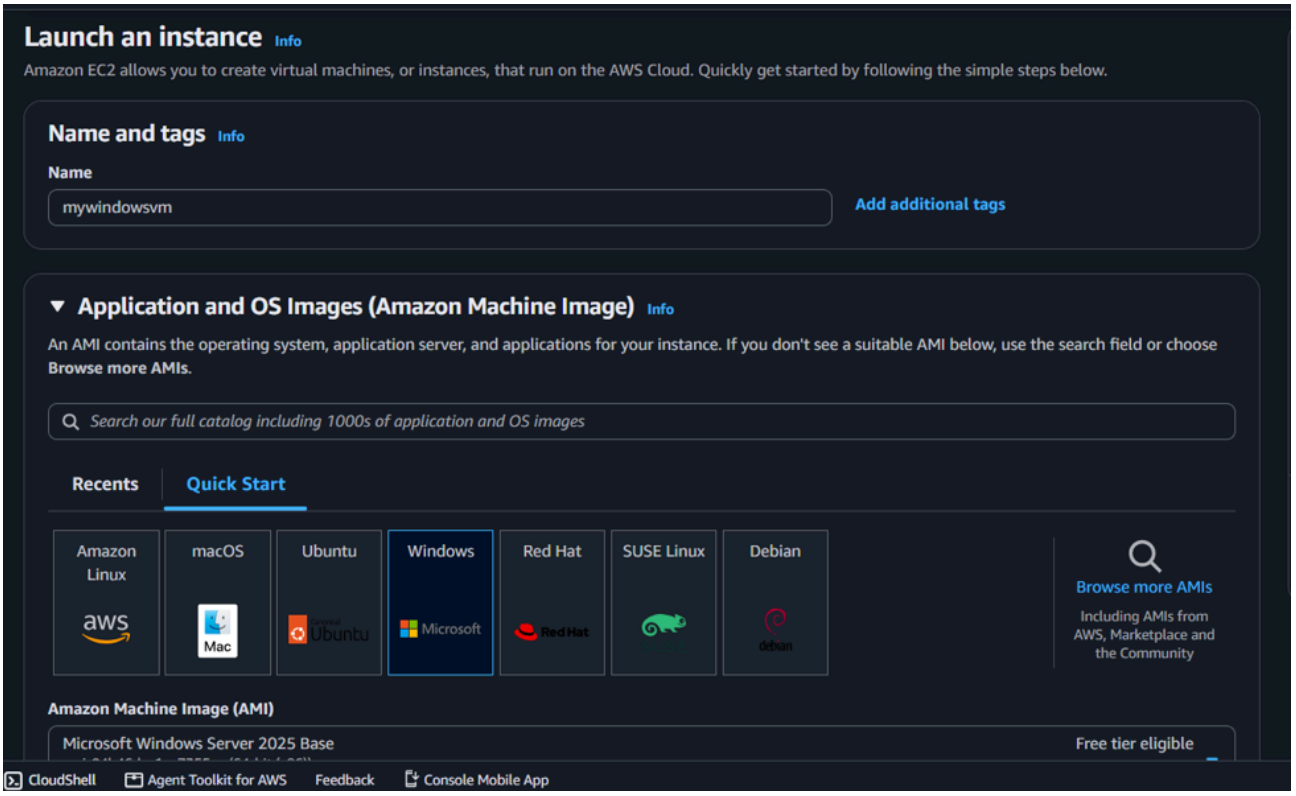
A new security group **launch-wizard-4** is created. **Allow RDP traffic** is ticked so that Remote Desktop can reach the instance, and HTTP traffic is also allowed. AWS warns that a source of 0.0.0.0/0 (Anywhere) lets every IP address in; it is safer to limit access to known IP addresses.

Step 5: Launch the instance



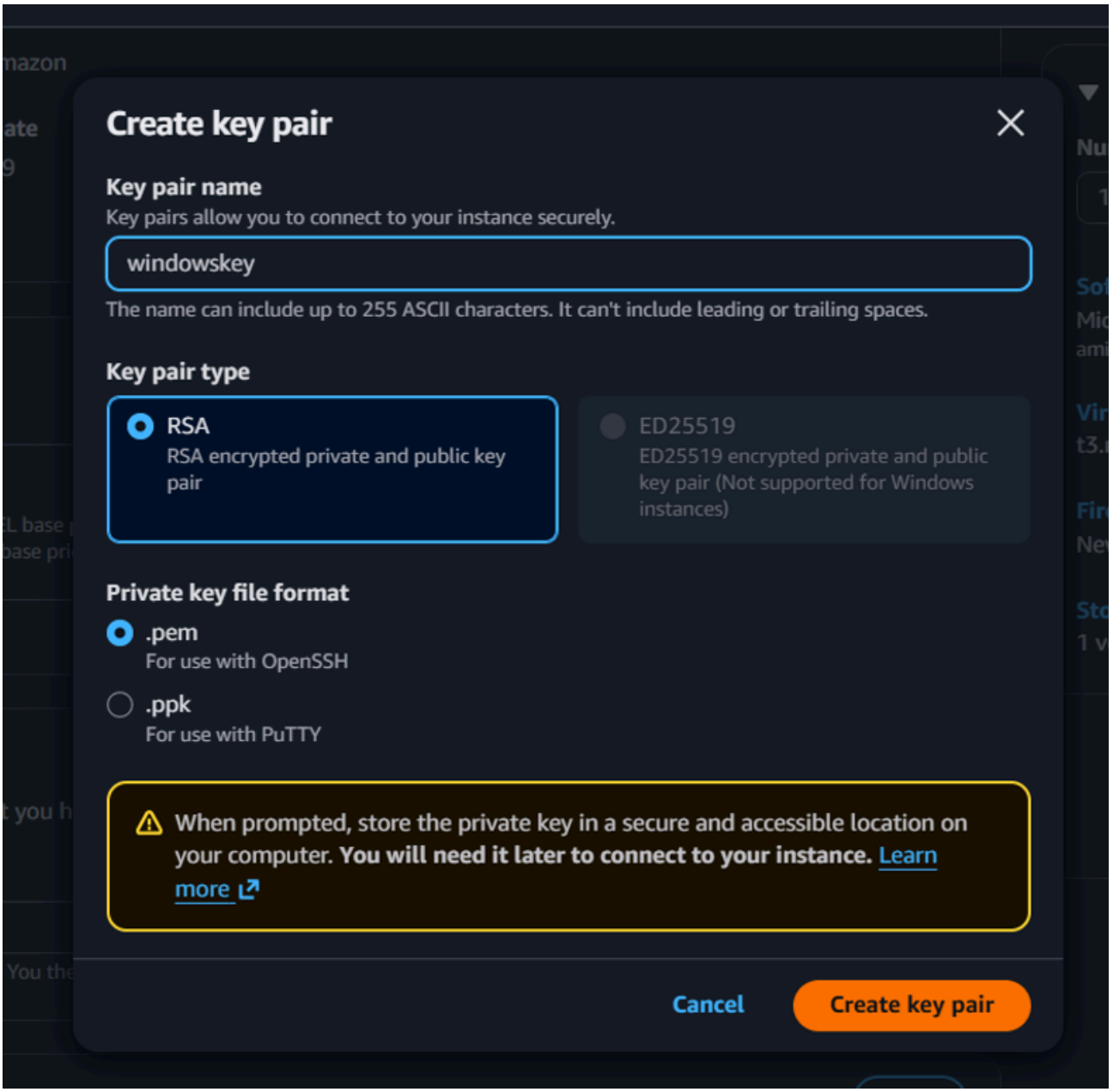
After **Launch instance** is clicked, AWS shows the progress of the launch. The page should stay open until it completes.

Step 6: Instance is running



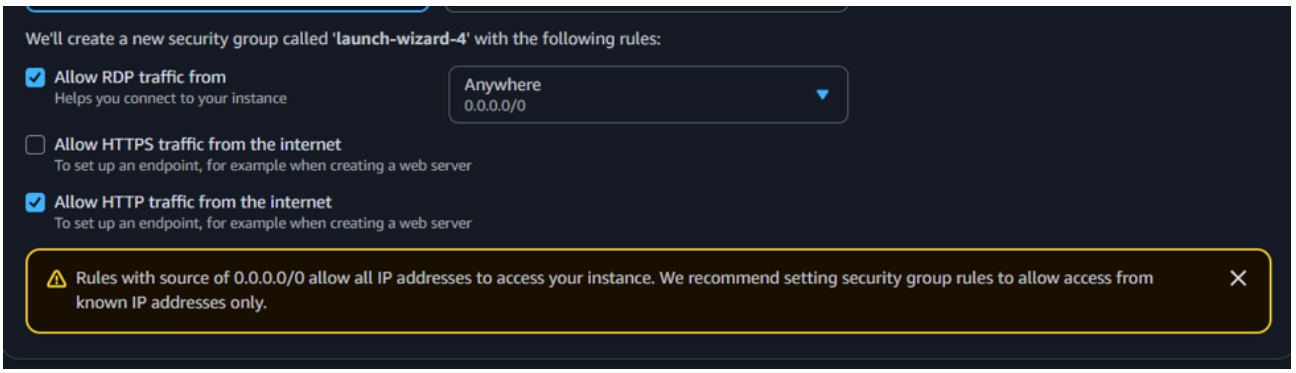
mywindowsvm now appears with the state **Running** and the status check Initializing. The **Details** tab shows its public IPv4 address, private IP address and public DNS name.

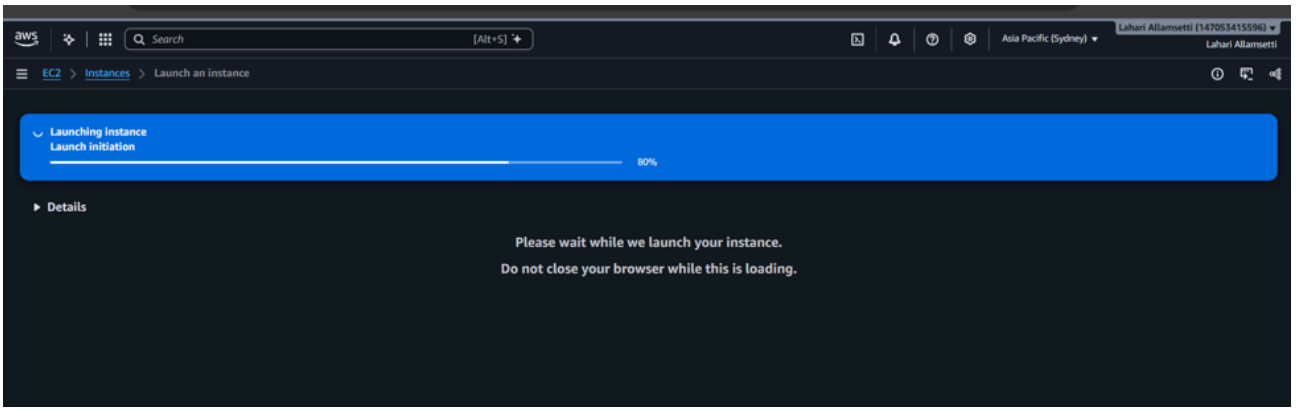
Step 7: Connect to the Windows instance

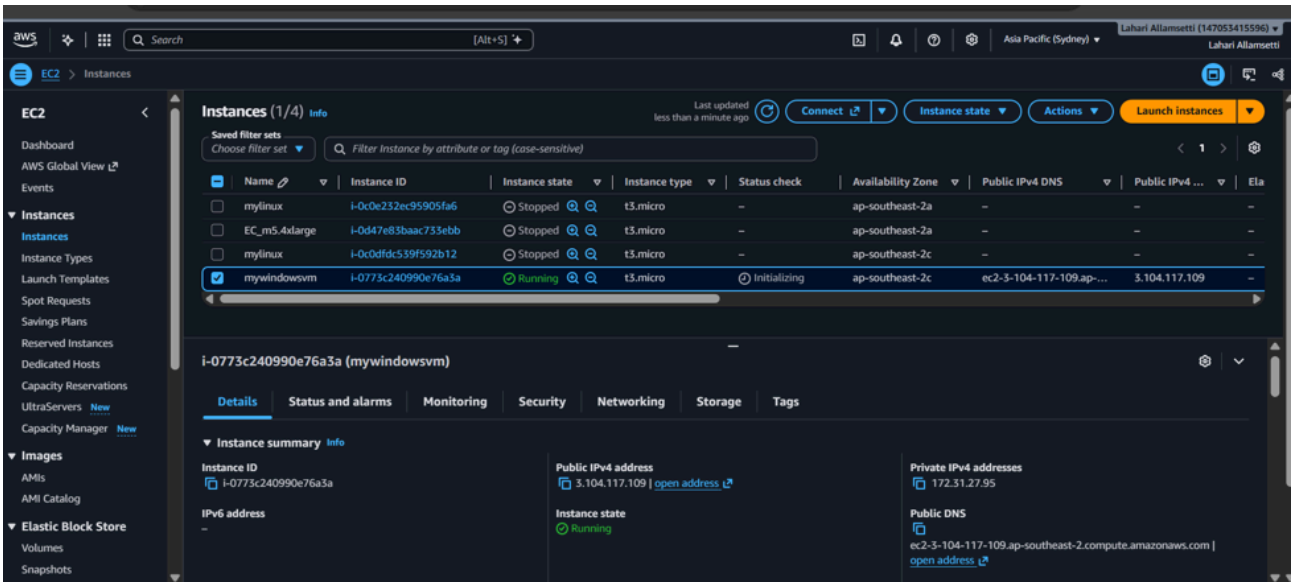


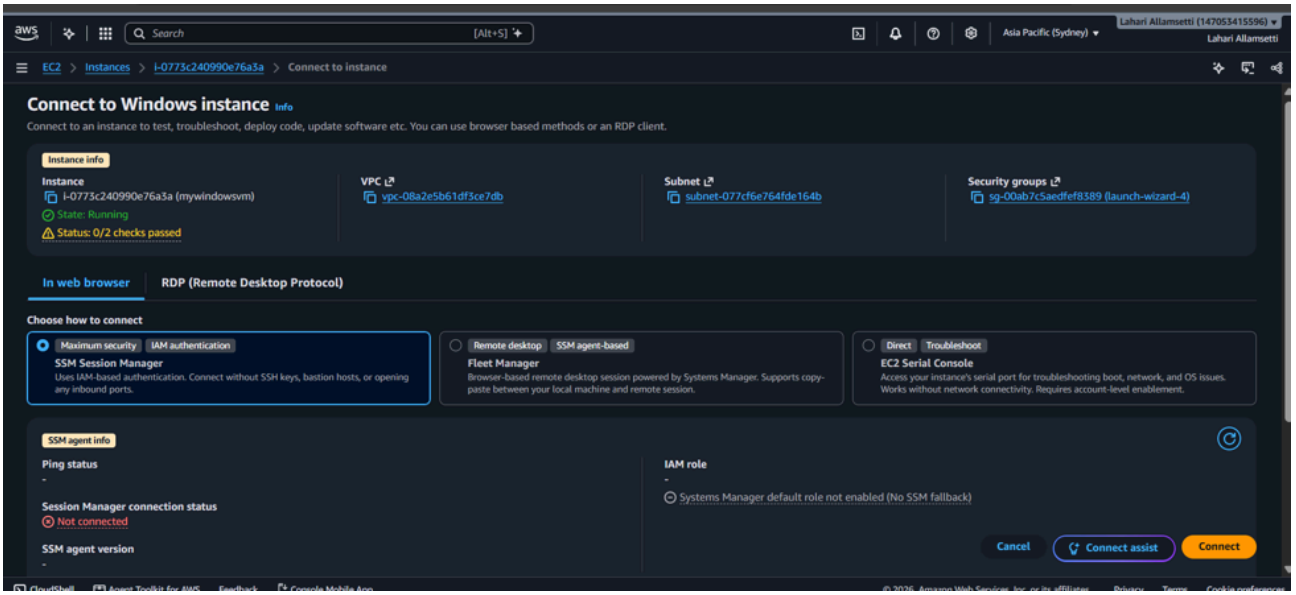
Clicking **Connect** opens this page with the connection options. The status is still **0/2 checks passed**, so it is necessary to wait until all the checks are passed before connecting.

Status checks passed









The status now reads **3/3 checks passed**, which means the instance is ready. The page offers **SSM Session Manager**, **Fleet Manager** and **EC2 Serial Console**. Session Manager is shown as Not connected because the SSM agent and IAM role are not set up, so Fleet Manager is used instead.

Step 8: Choose Fleet Manager

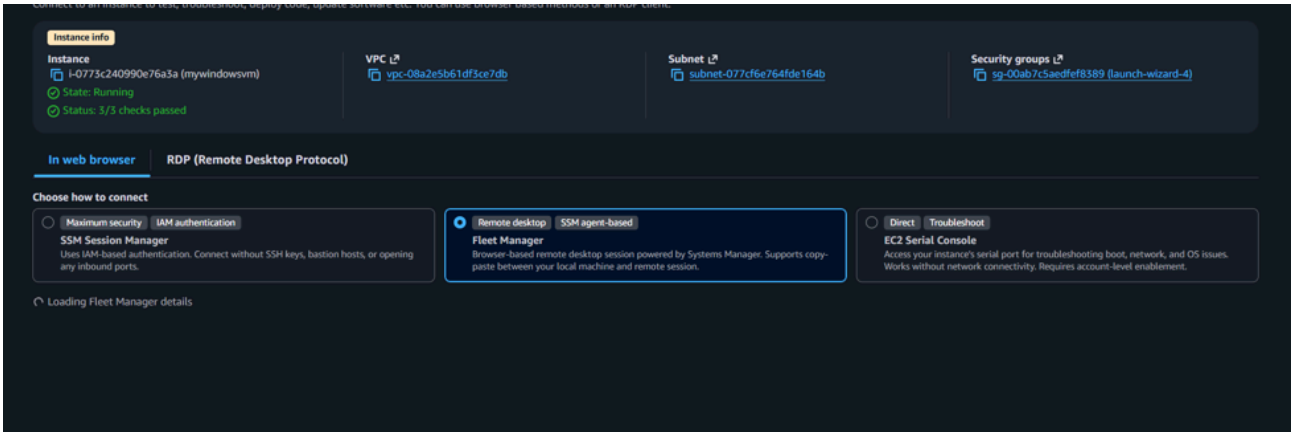


Fig: Fleet Manager Remote Desktop selected

Fleet Manager (Remote desktop) gives a browser-based remote desktop session. The page loads the Fleet Manager details after it is selected.

Step 9: How to connect

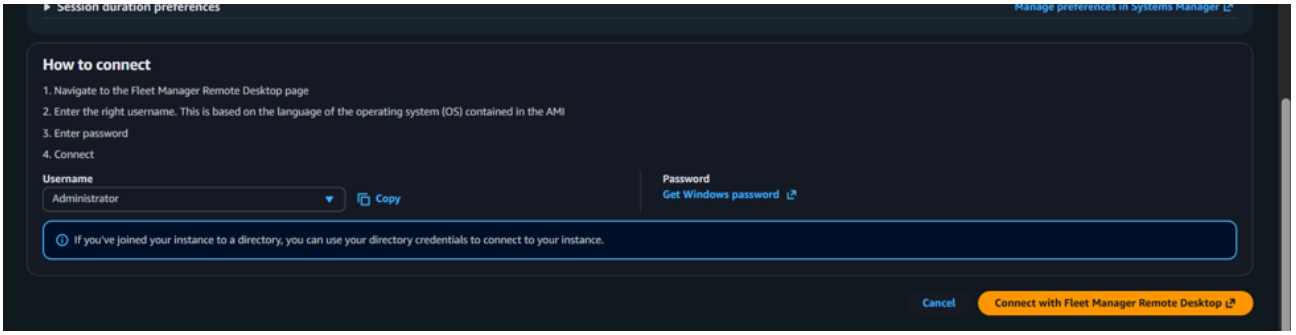
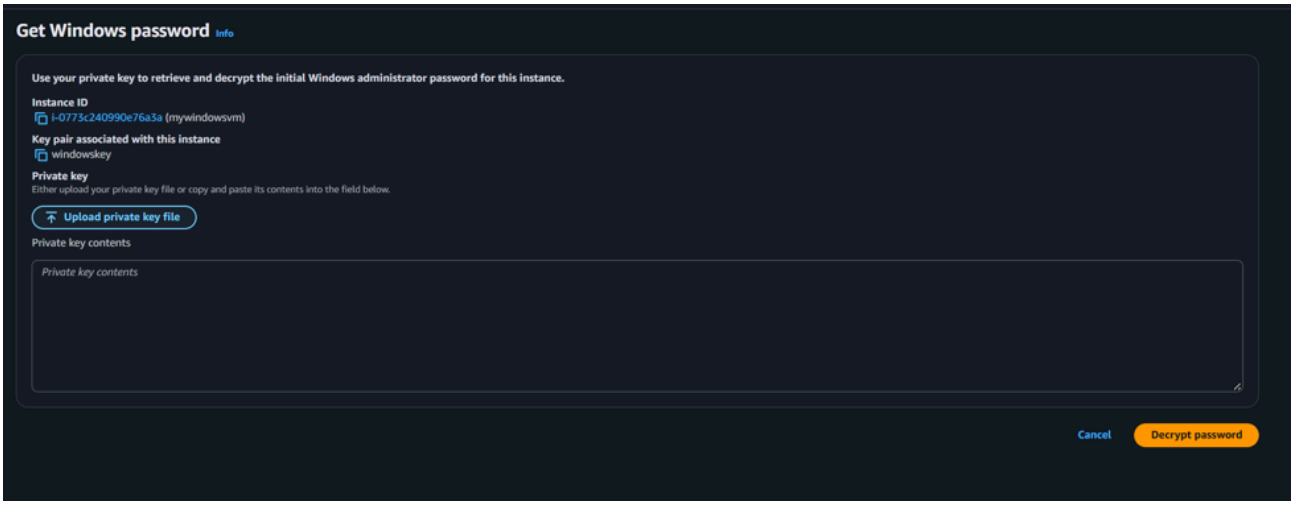


Fig: Fleet Manager – username and password

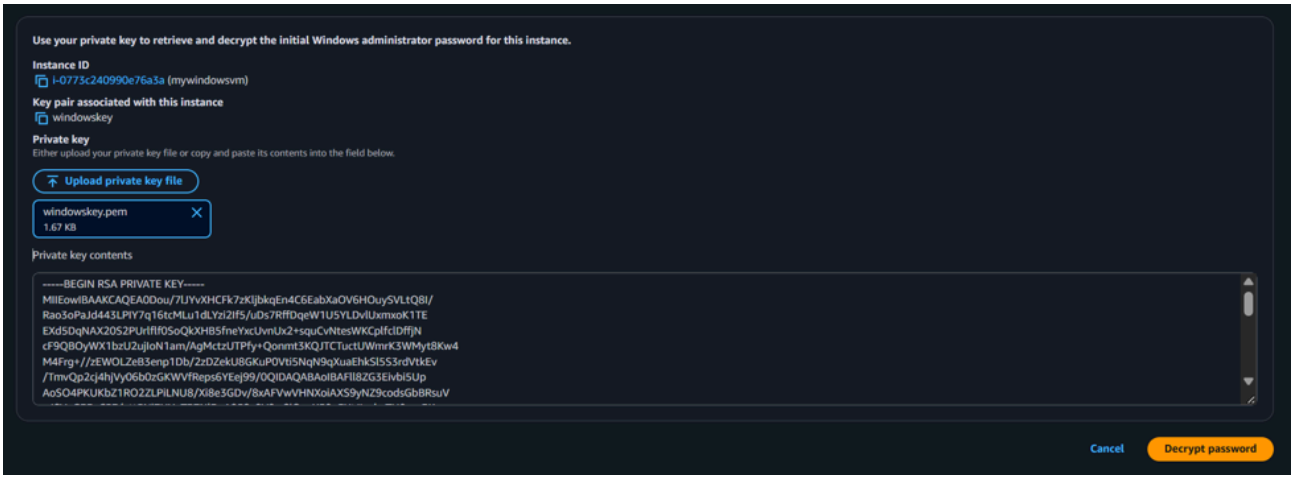
The instructions say to enter the right user name, which is **Administrator** for Windows, then to enter the password and connect. The password is obtained with **Get Windows password**.

Step 10: Get the Windows password



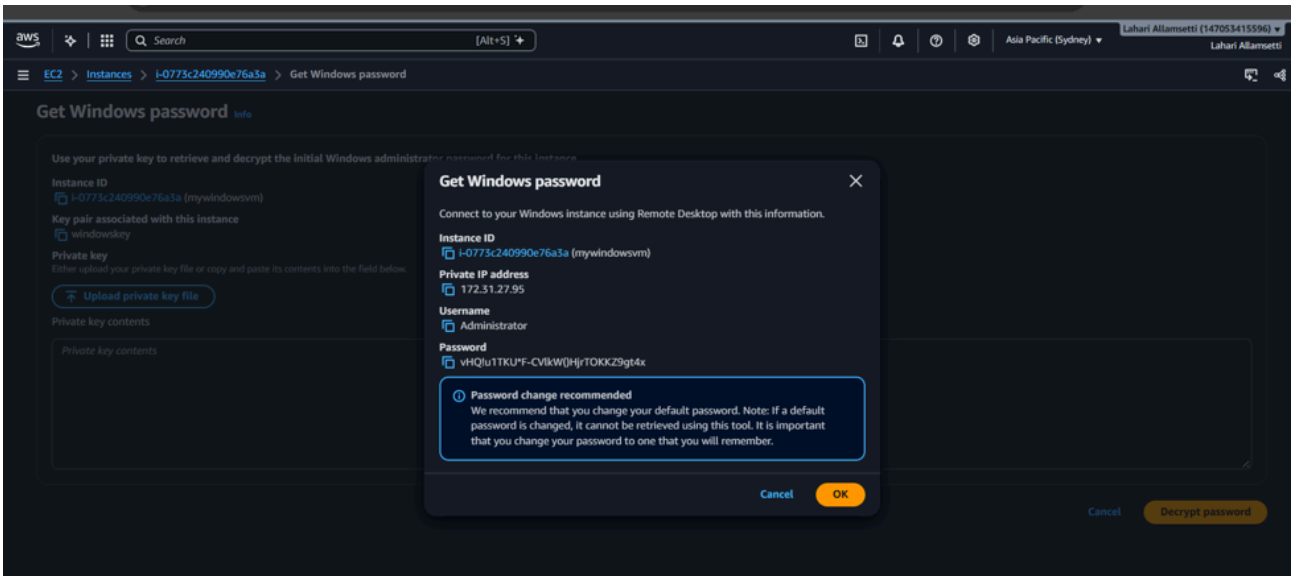
The password is protected with the key pair. The private key is supplied either with **Upload private key file** or by pasting its contents, and then **Decrypt password** is clicked.

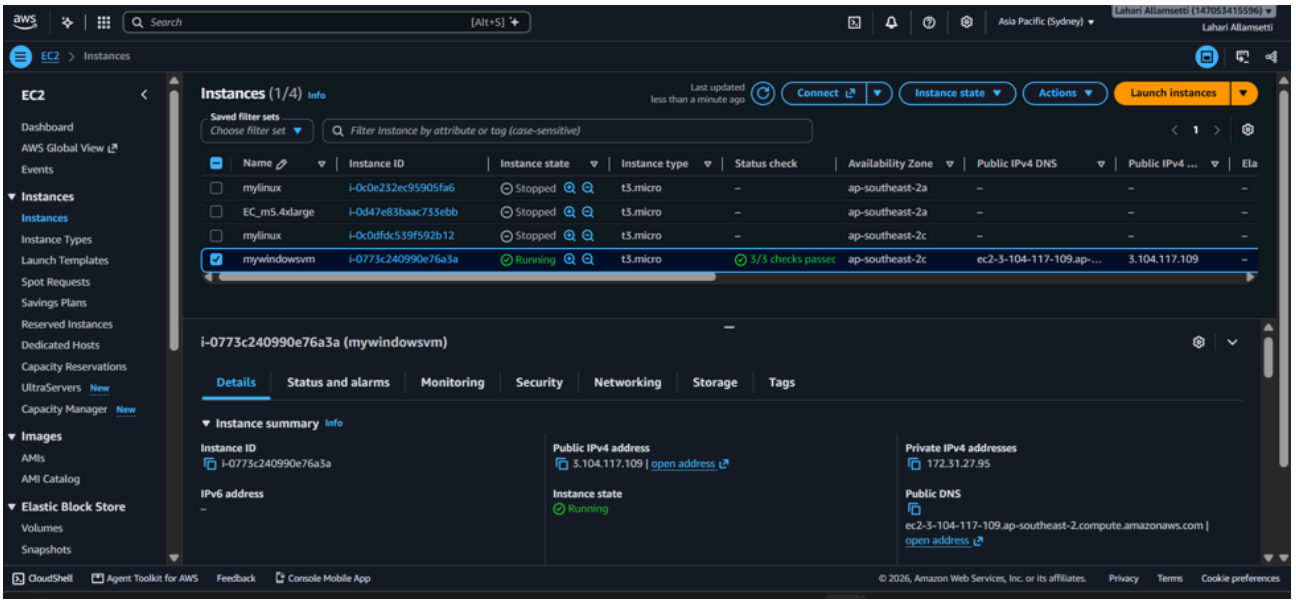
Upload the private key



The saved **windowskey.pem** file (1.67 KB) is uploaded, and its contents appear in the Private key contents box. **Decrypt password** is then clicked.

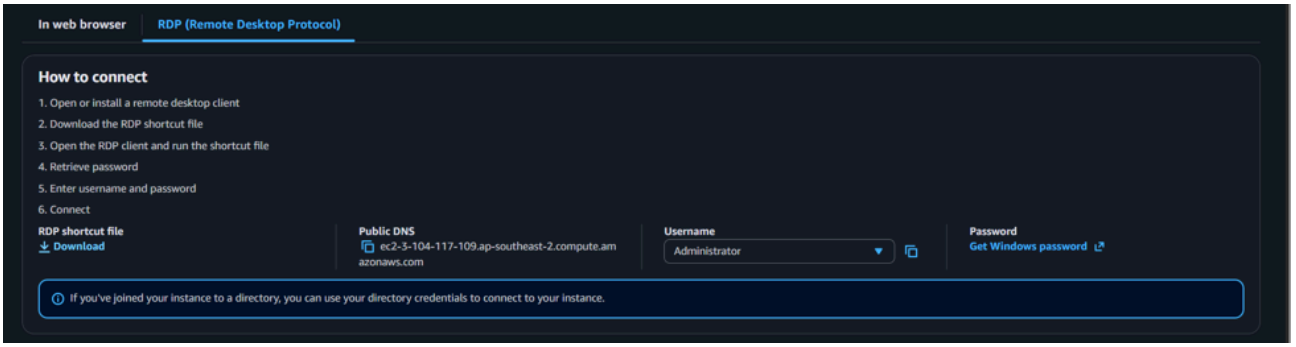
Password decrypted

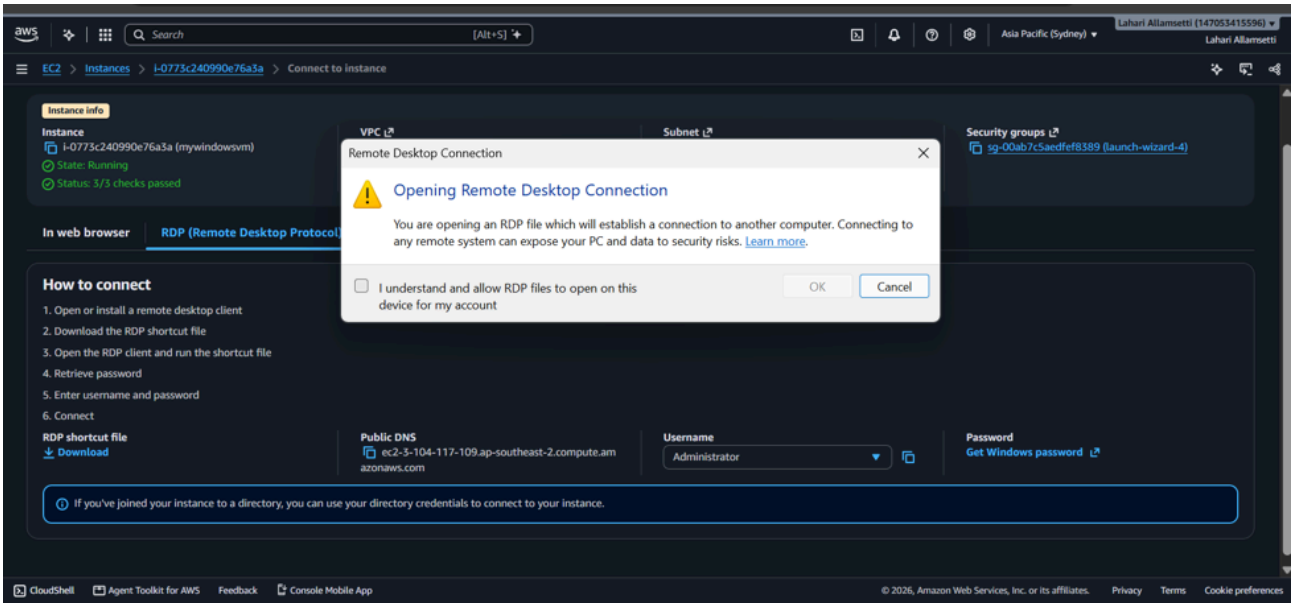




AWS shows the connection details: the instance ID, the private IP address, the user name **Administrator** and the decrypted password. AWS recommends changing this default password after the first login to one that is easy to remember. This password should be kept private.

Step 11: Connect using RDP





Back on the Instances page, **mywindowsvm** is selected and shows **3/3 checks passed** with the

state Running, so it is ready for a Remote Desktop connection.

RDP tab

Fig: How to connect using RDP

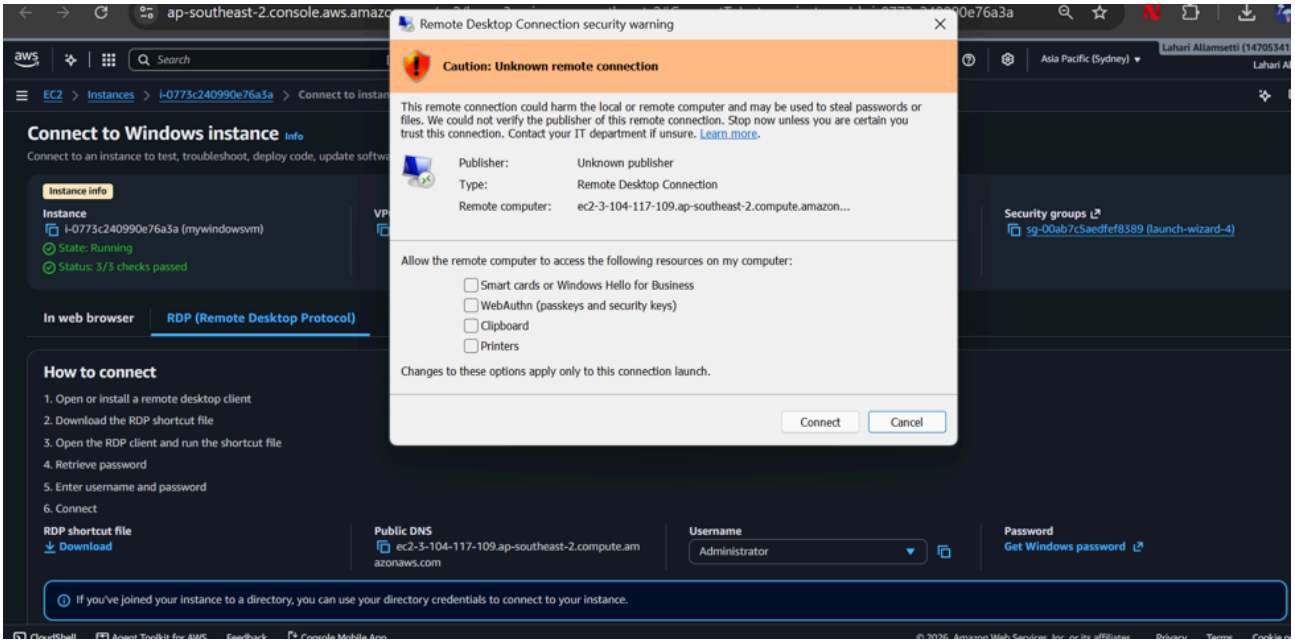
In the **RDP (Remote Desktop Protocol)** tab the steps are: install a remote desktop client, download the RDP shortcut file, run it, retrieve the password, enter the user name and password, and connect. The page gives the public DNS of the instance and the user name **Administrator**.

Open the RDP file

Fig: Opening Remote Desktop Connection dialog

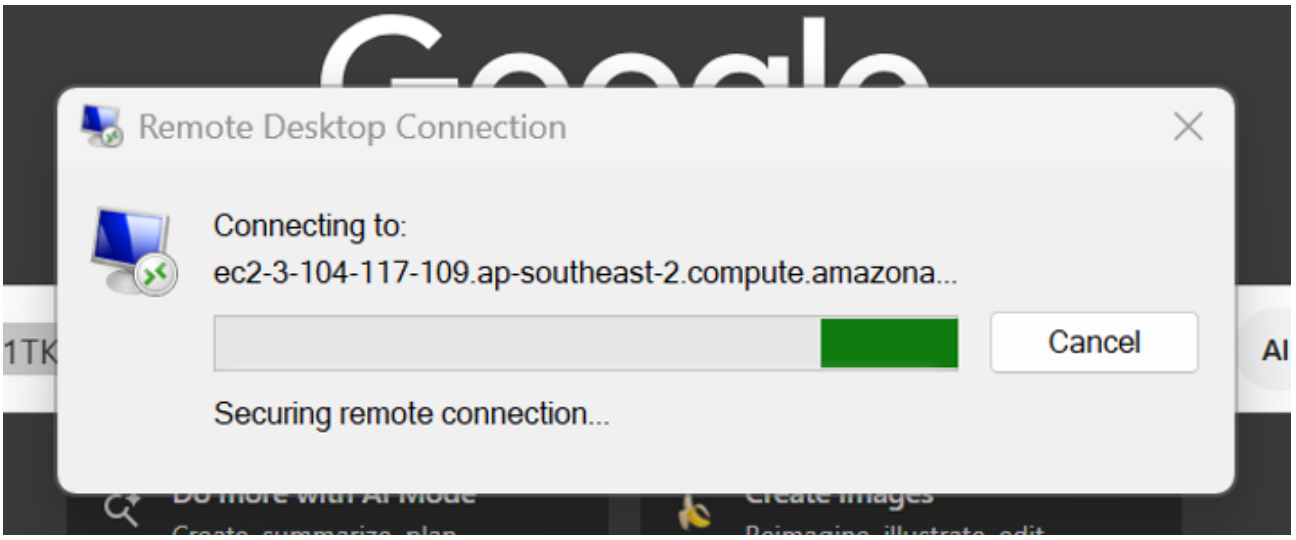
After the RDP shortcut file is downloaded and opened, Windows warns that it is opening an RDP file that connects to another computer. The box **I understand and allow RDP files to open on this device** is ticked and **OK** is clicked.

Security warning



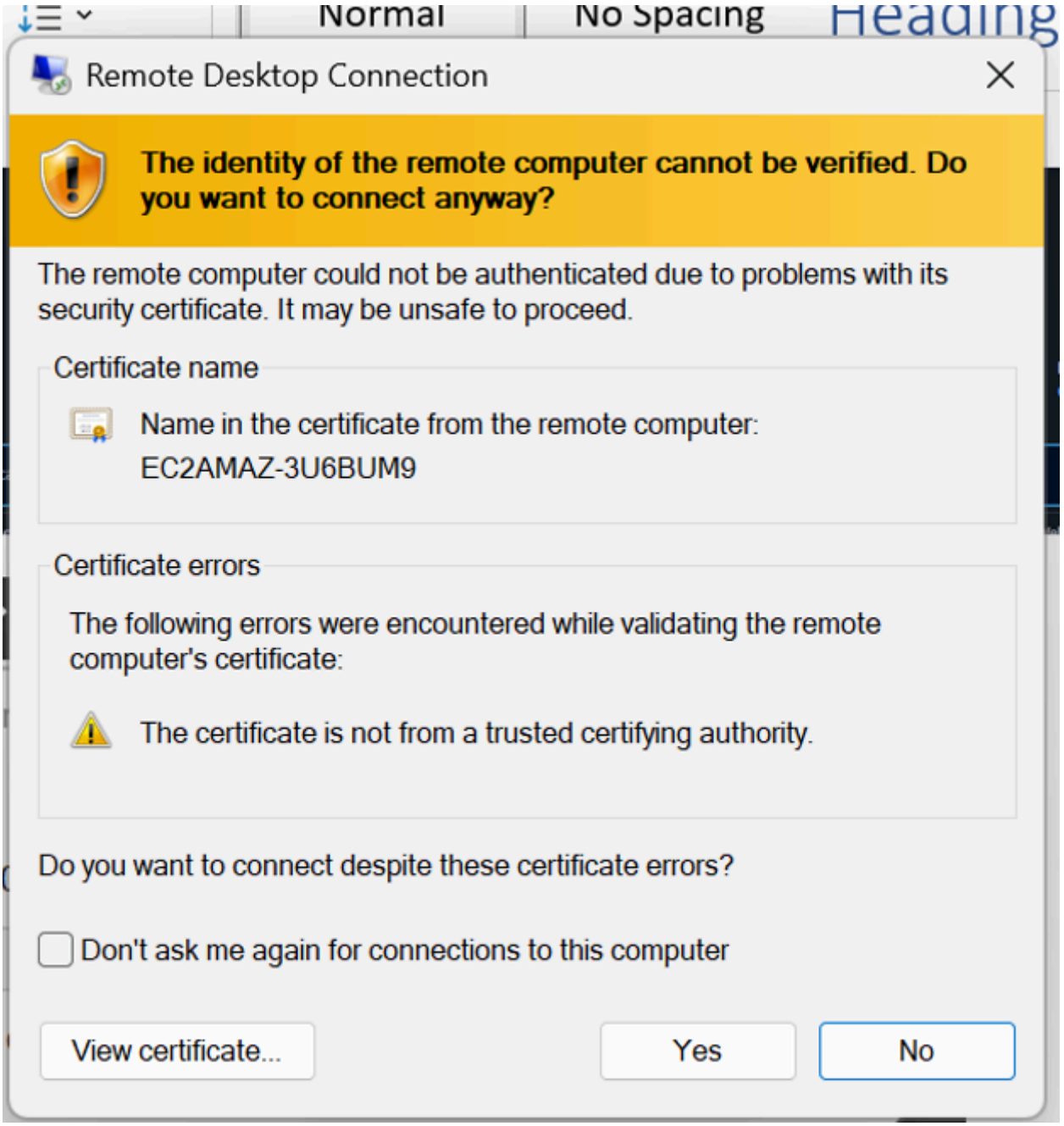
Windows shows a caution because the publisher of the connection is unknown. As the connection is to our own instance, **Connect** is clicked. The options for redirecting resources (clipboard, printers, smart cards) are left as needed.

Connecting



The Remote Desktop client now connects to the public DNS of the instance and shows the message Securing remote connection.

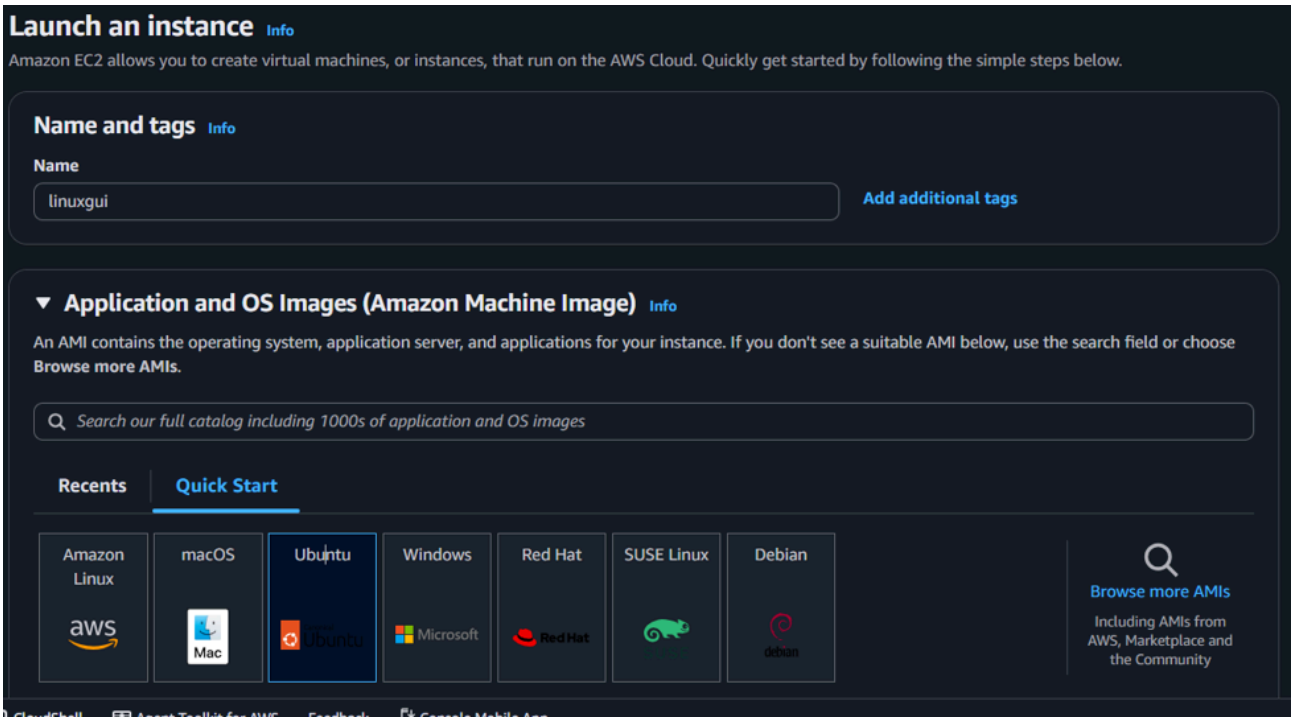
Certificate warning



The server certificate was issued by the instance itself and is not from a trusted certifying authority, so Windows asks whether to connect anyway. Because the server is our own, **Yes** is clicked (the box Don't ask me again is optional), and the Windows desktop of the instance opens after logging in.

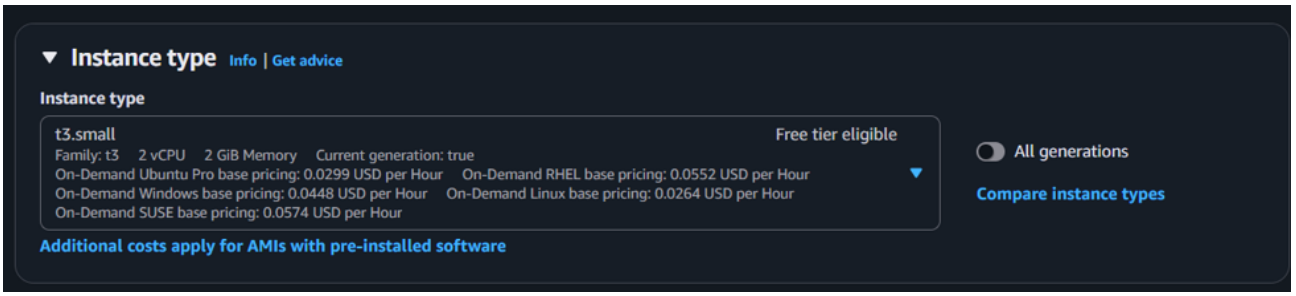
2B. Linux (Ubuntu) instance with a graphical desktop

Step 1: Name and select Ubuntu



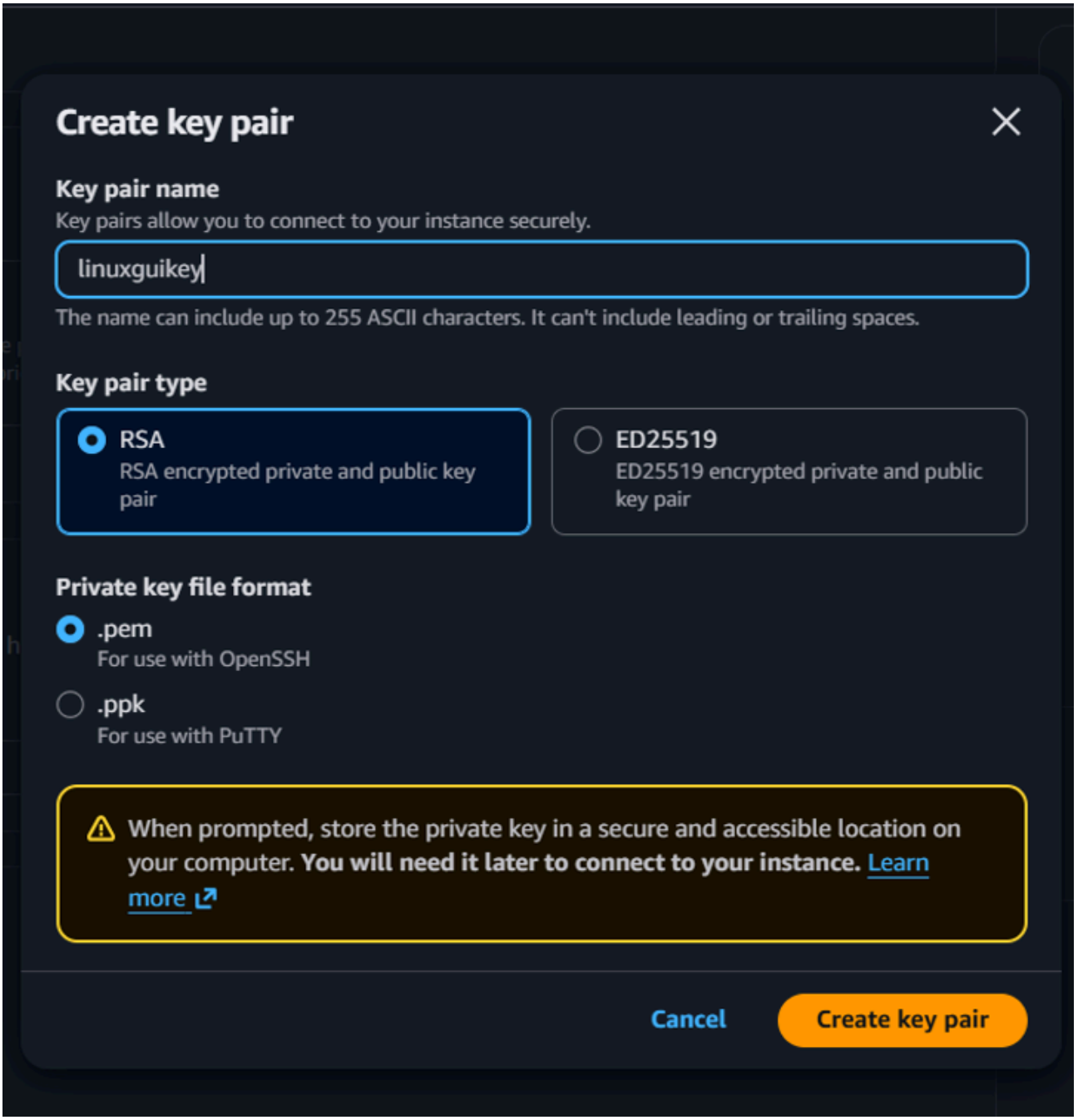
A new instance is named **linuxgui**, and the **Ubuntu** tile is chosen in the Quick Start list as the operating system.

Step 2: Instance type



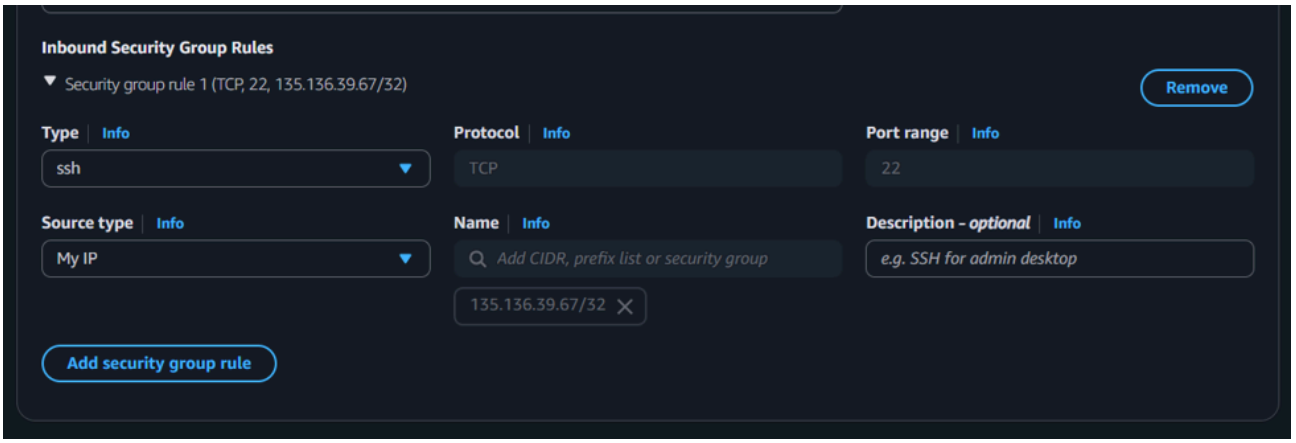
A larger type, **t3.small** (2 vCPU, 2 GiB memory), is selected because a graphical desktop needs more memory than the basic t3.micro.

Step 3: Create a key pair



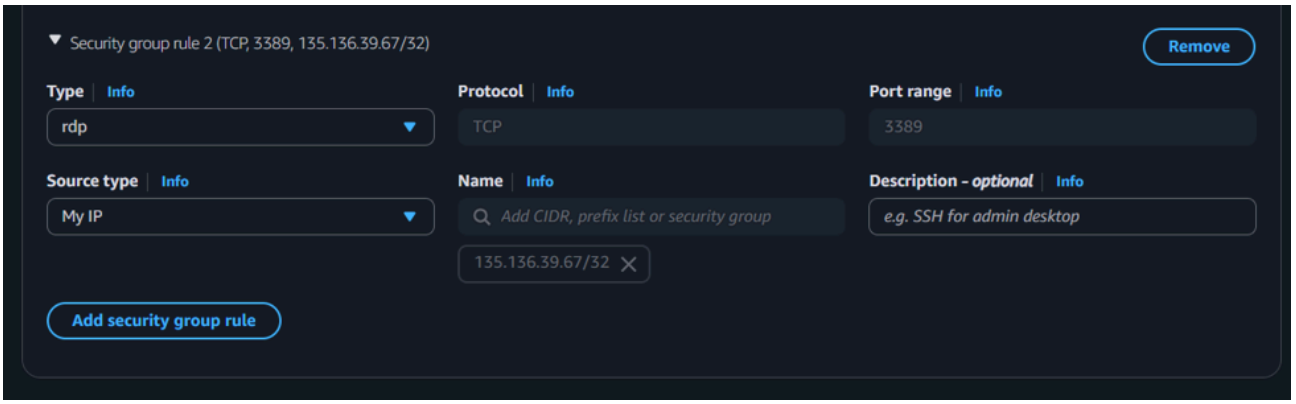
A key pair named **linuxguikey** is created with the **RSA** type and the **.pem** format, and the file is saved safely.

Step 4: Security group rules



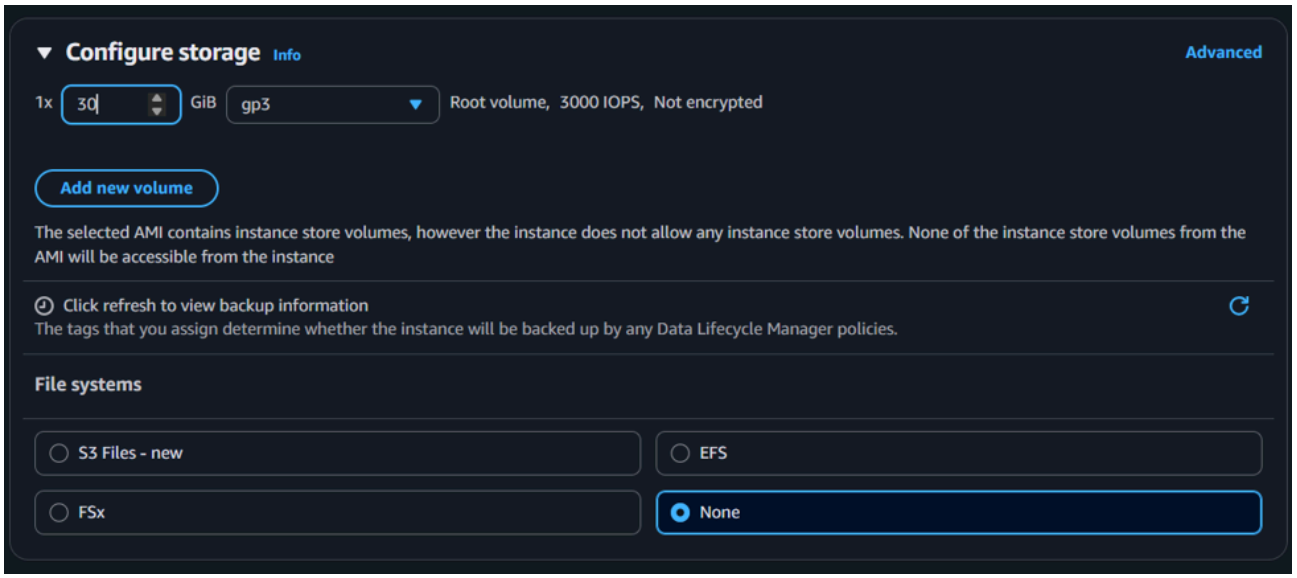
In the inbound security group rules, rule 1 allows **SSH** (TCP, port 22) with the source type **My IP**, so that only the user's own IP address can log in over SSH.

Add a second rule



Add security group rule is used to add rule 2, which allows **RDP** (TCP, port 3389) from **My IP**. This opens the port needed for the Remote Desktop connection to the Ubuntu desktop.

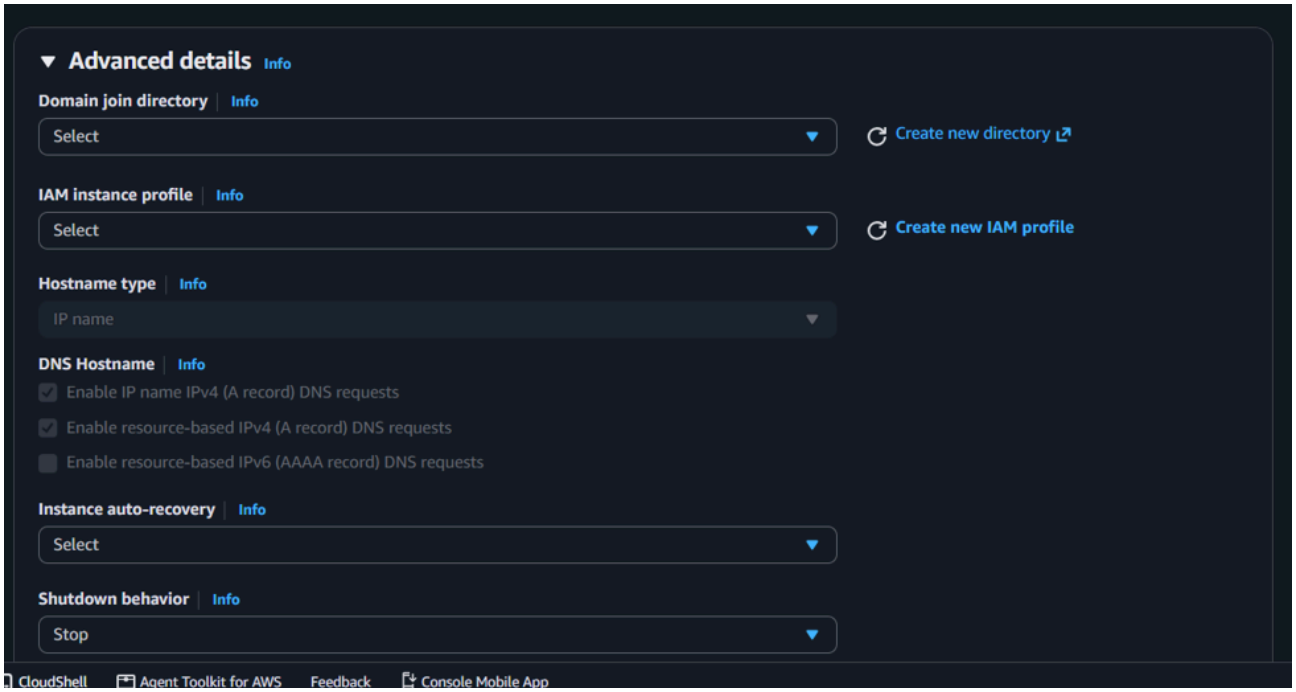
Step 5: Configure storage



The root volume is set to **30 GiB** of **gp3** storage (3000 IOPS, not encrypted) so that there is enough

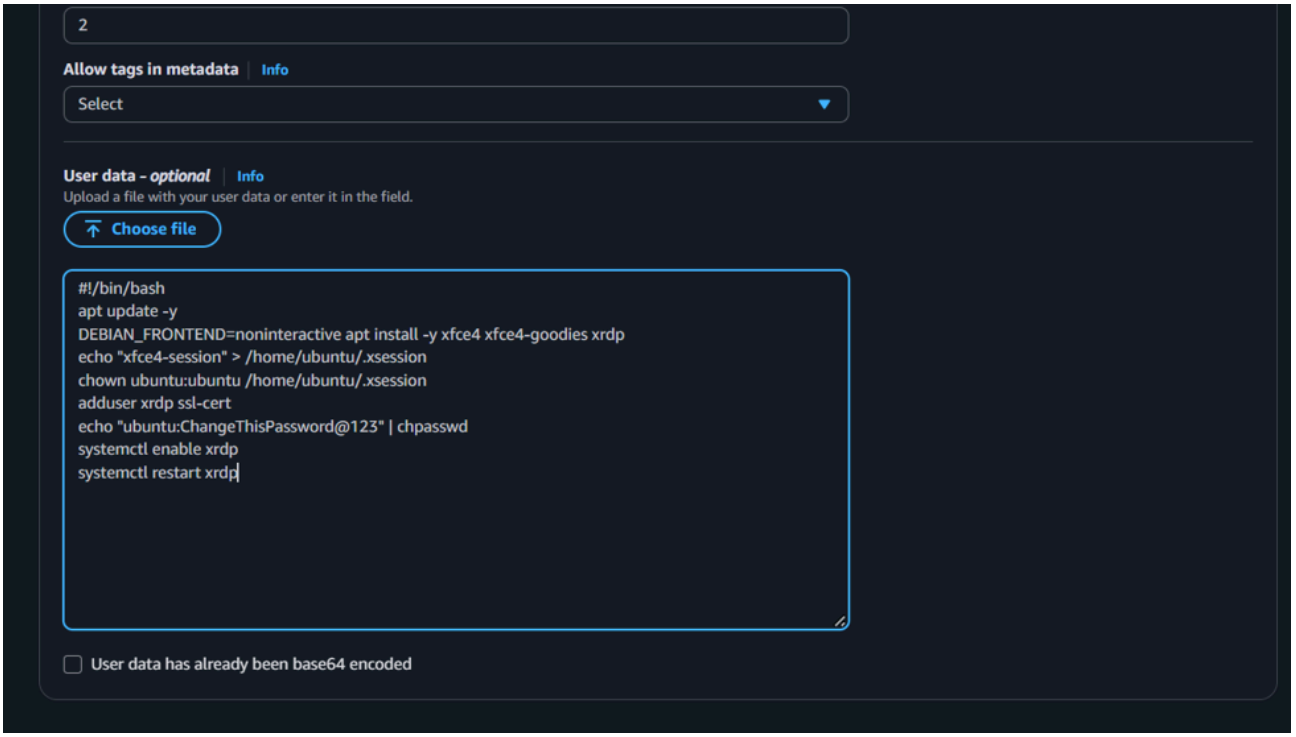
space for the desktop software. No extra file system (S3 Files, EFS or FSx) is attached, so **None** is kept.

Step 6: Advanced details



The **Advanced details** section holds options such as the domain join directory, IAM instance profile, hostname type, DNS hostname settings, instance auto-recovery and shutdown behaviour (**Stop**). The defaults are kept here. Scrolling further down gives the **User data** box.

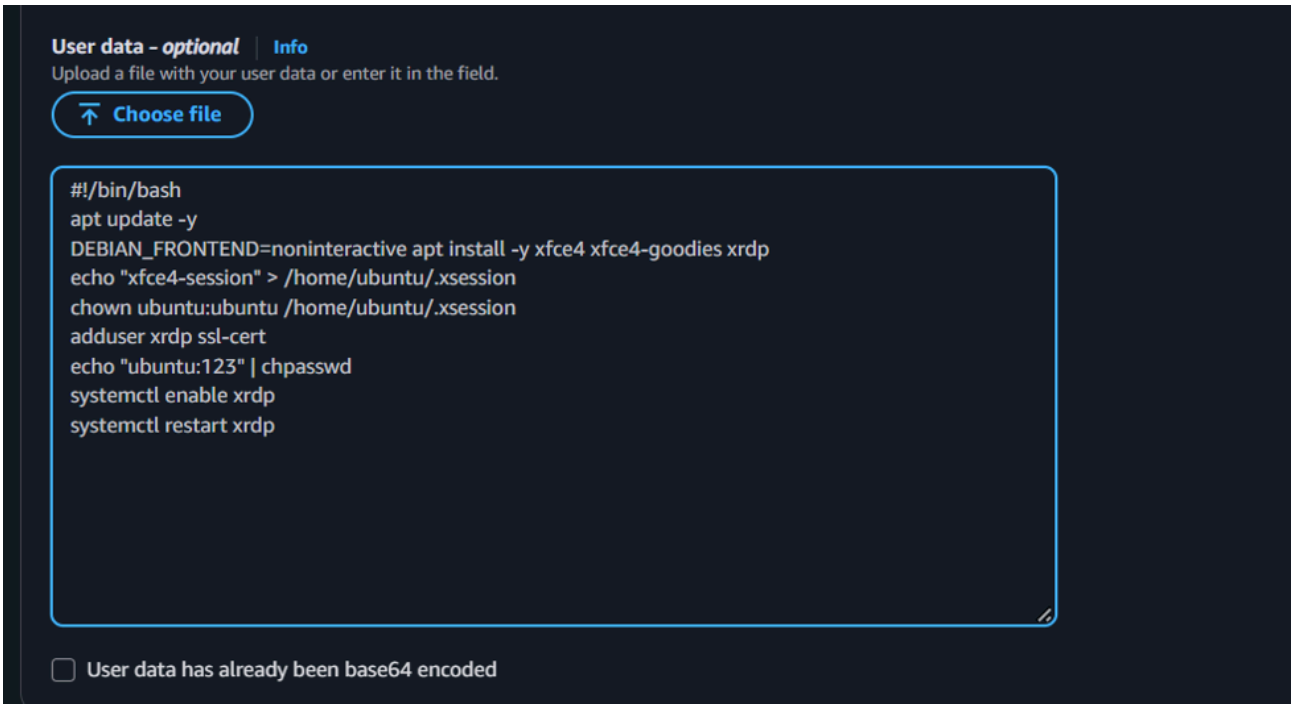
User data script (first version)



User data is a script that runs automatically when the instance starts for the first time. Here a bash script

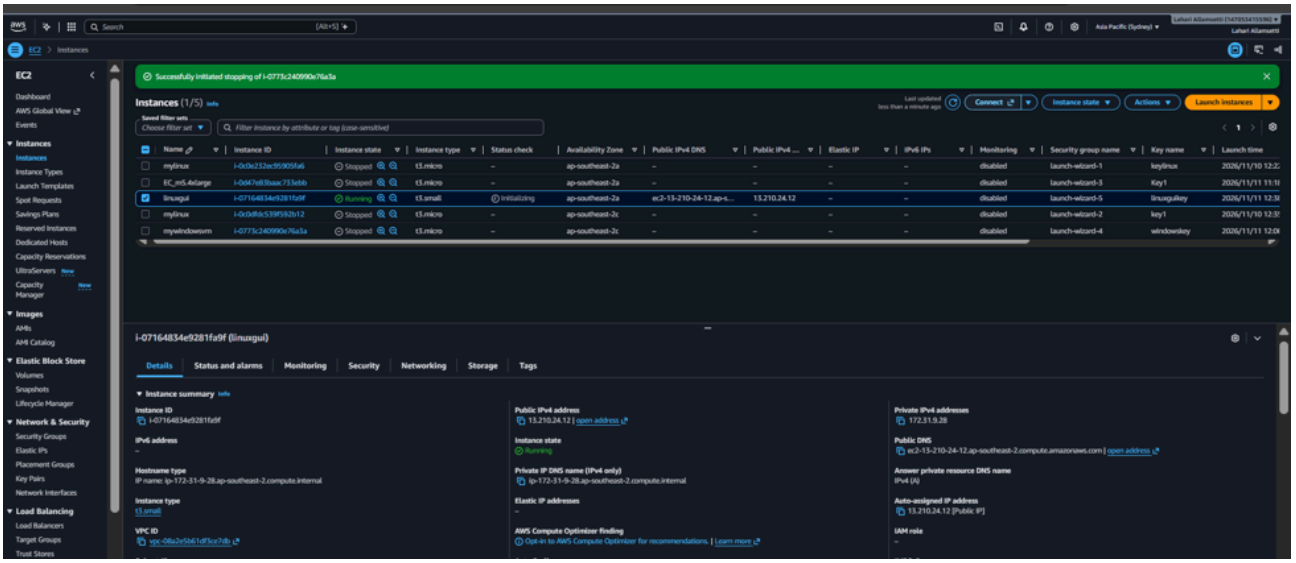
updates the packages, installs the **xfce4** desktop and the **xrdp** Remote Desktop server, creates the desktop session file for the **ubuntu** user, sets a password for that user, and enables and restarts the **xrdp** service.

User data script (final version)



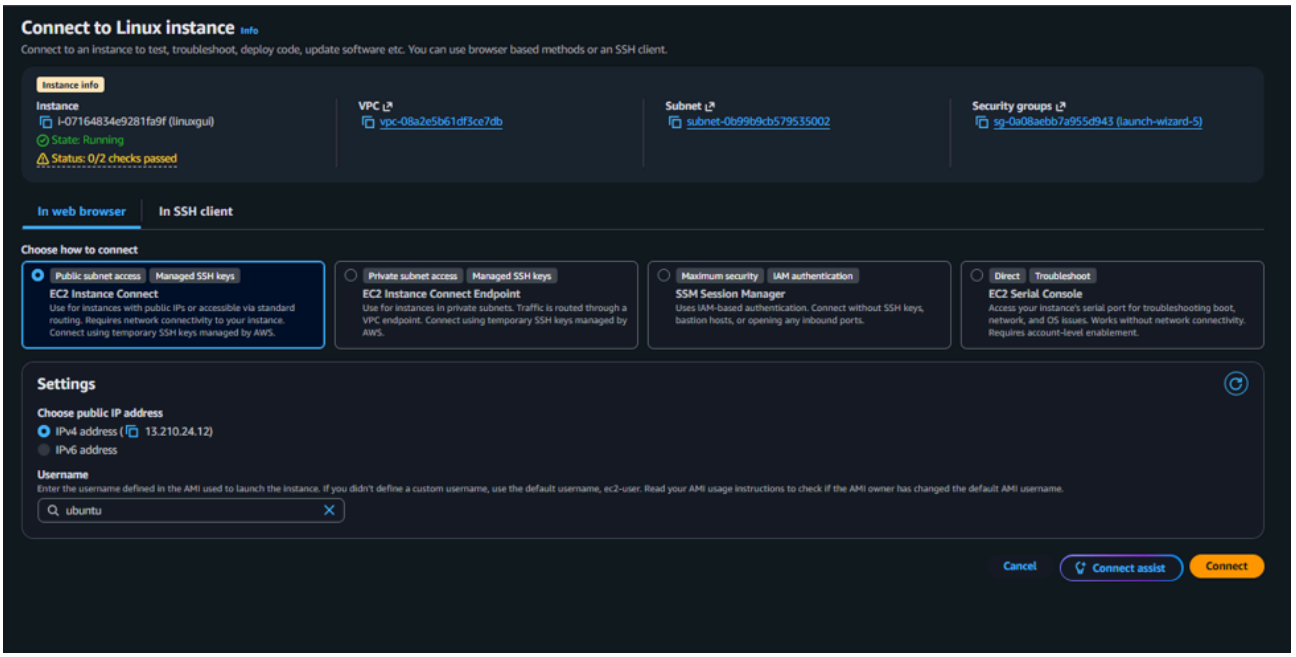
The final script has the same steps with the ubuntu user's password changed to one that will be used for the RDP login. User data has already been base64 encoded is left unticked since the text is plain. A strong password should be used for real servers. Then **Launch instance** is clicked.

Step 7: Launch



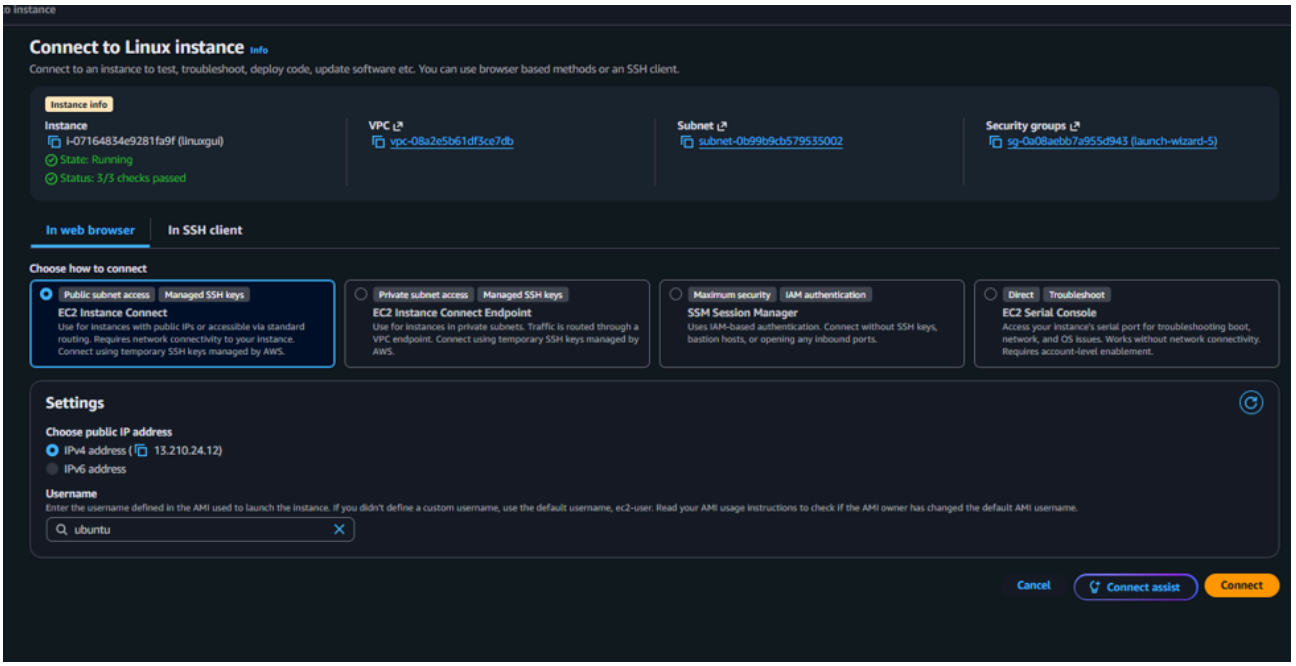
The launch progress bar is shown while AWS creates the instance.

Instance running



linuxgui (t3.small) is listed with the state **Running**, and its public IPv4 address and key name **linuxguikey** are visible. It first shows Initializing while the checks run.

Step 8: Connect to the Linux instance

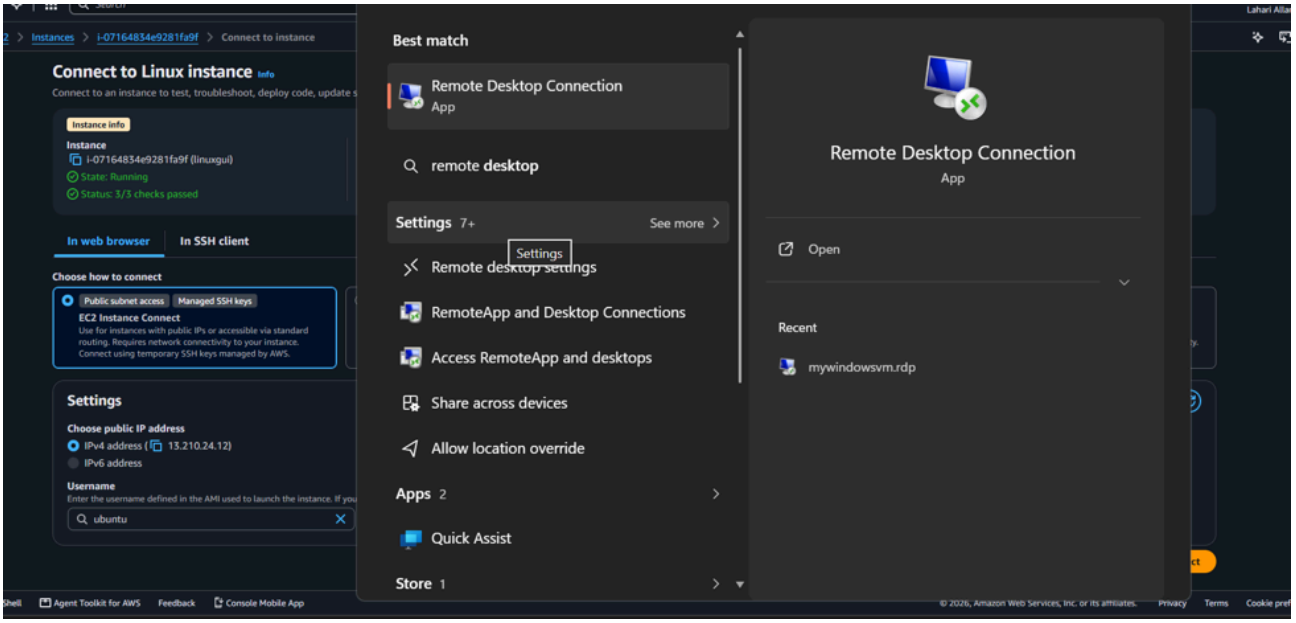


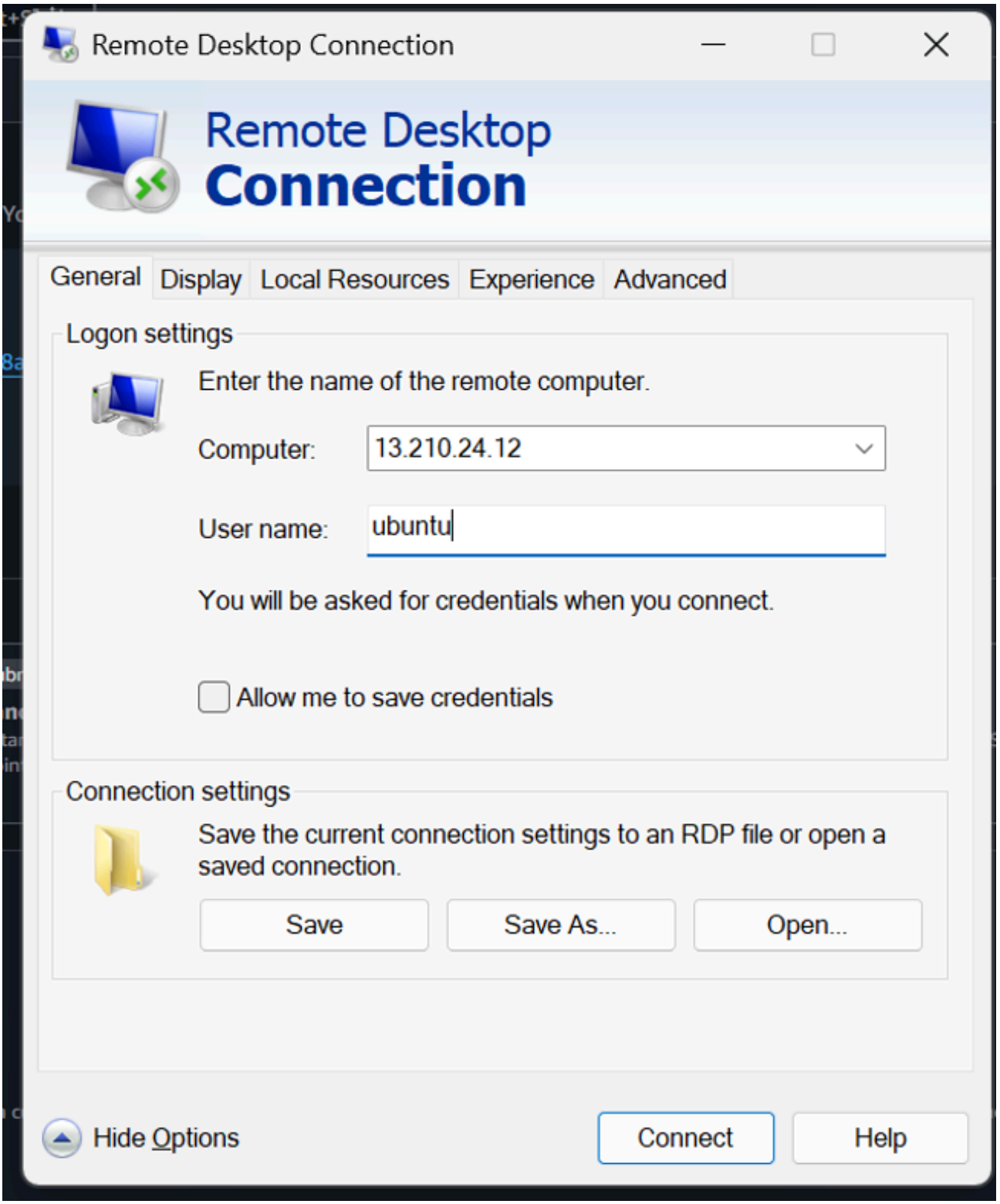
The **Connect to Linux instance** page offers **EC2 Instance Connect** and other methods, with the default user name **ubuntu**. The status is still **0/2 checks passed**, so it is necessary to wait.

Status checks passed

When the status reads **3/3 checks passed**, the instance is ready. Since the xrdp desktop was installed through user data, the connection is made from the Windows Remote Desktop client using the instance's public IPv4 address.

Step 9: Open Remote Desktop Connection





On the local Windows computer, **remote desktop** is typed in the Start menu and the **Remote Desktop Connection** app is opened.

Enter the connection details

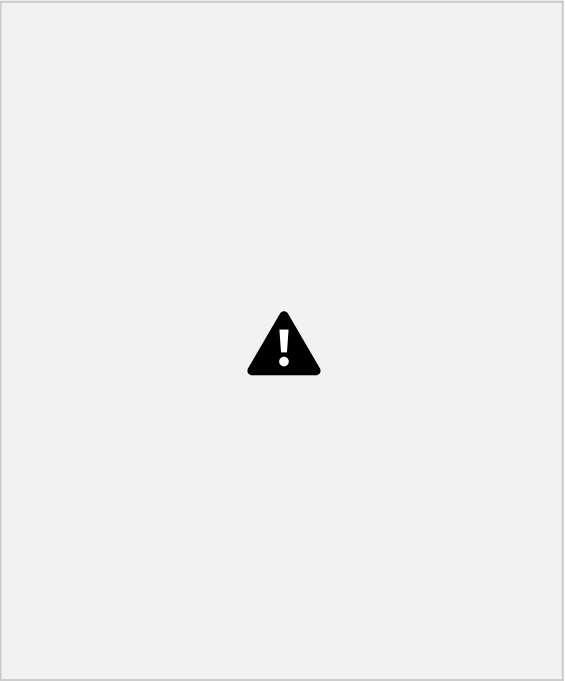
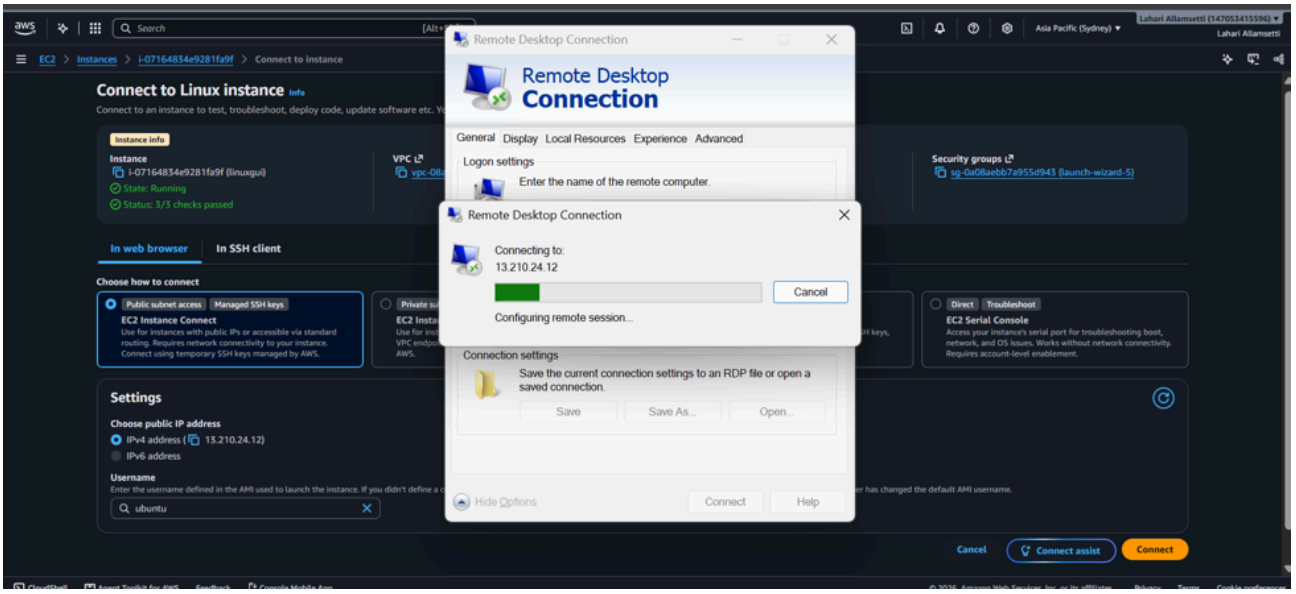


Fig: Remote Desktop Connection window

In the **Computer** field the public IPv4 address of the instance (**13.210.24.12**) is entered, and the user name is **ubuntu**. **Connect** is clicked, and the password set in the user data script is entered when prompted.

Connecting



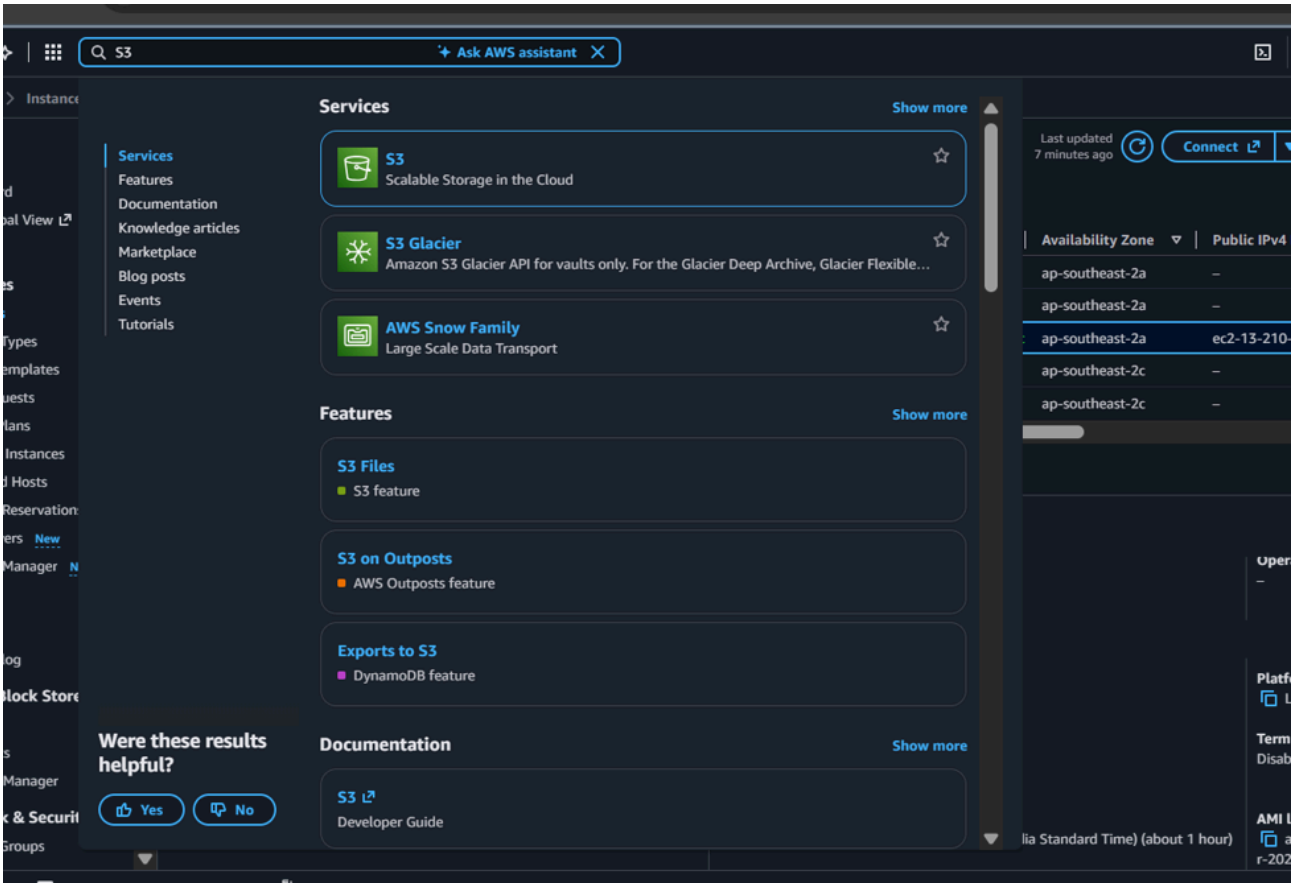
The client shows Configuring remote session while it connects. When it completes, the graphical Ubuntu (xfce4) desktop of the cloud instance appears on the local screen.

Result: A Windows Server instance (connected through Fleet Manager and RDP using a decrypted Administrator password) and an Ubuntu instance with an xfce4 desktop (connected through xrdp using Remote Desktop) were created successfully on Amazon EC2.

PART 3: CREATING AN S3 BUCKET

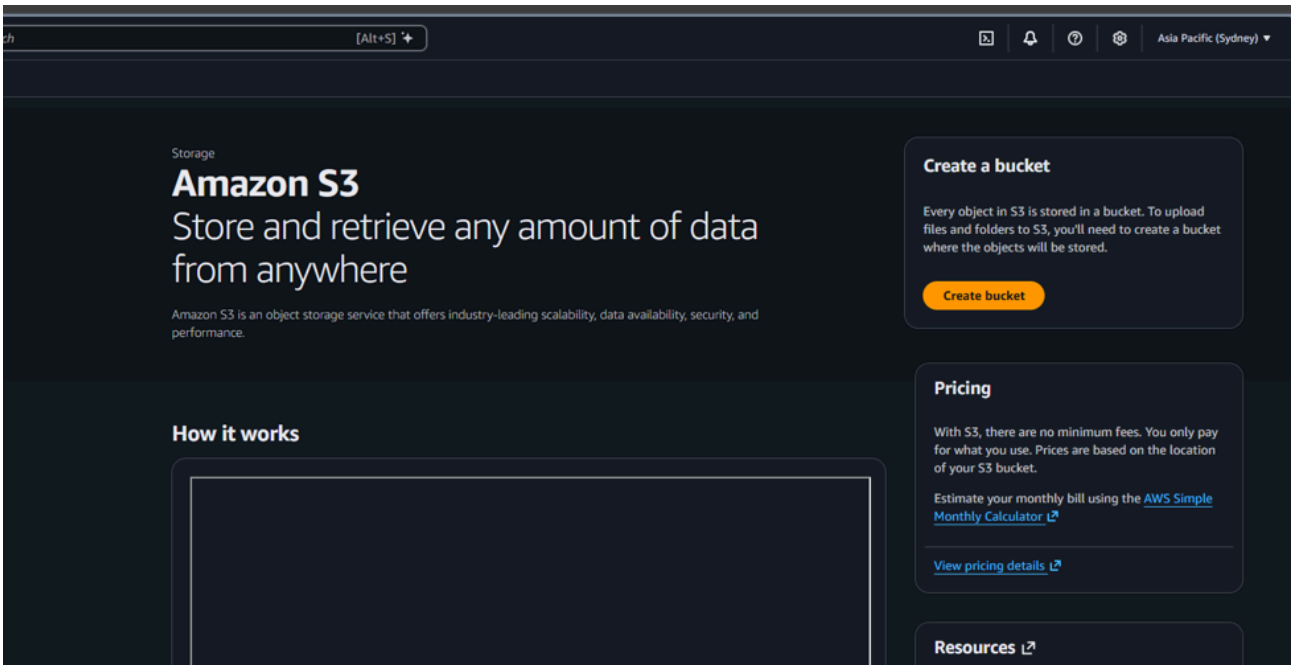
Amazon S3 (Simple Storage Service) is an object-storage service for storing and retrieving any amount of data from anywhere. Files (objects) in S3 are kept inside containers called **buckets**, so a bucket is a space in which the data is stored.

Step 1: Search for S3



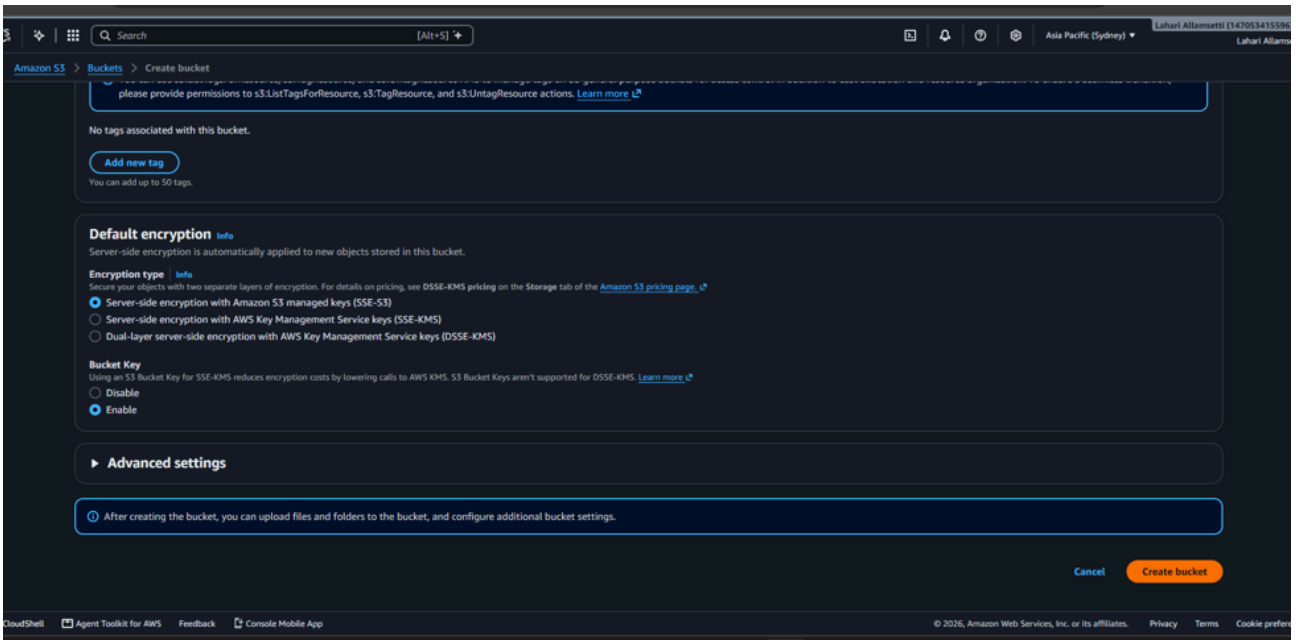
S3 is typed in the search bar of the AWS console. The results list **S3 – Scalable Storage in the Cloud** under Services, which is clicked to open the service.

Step 2: Create a bucket



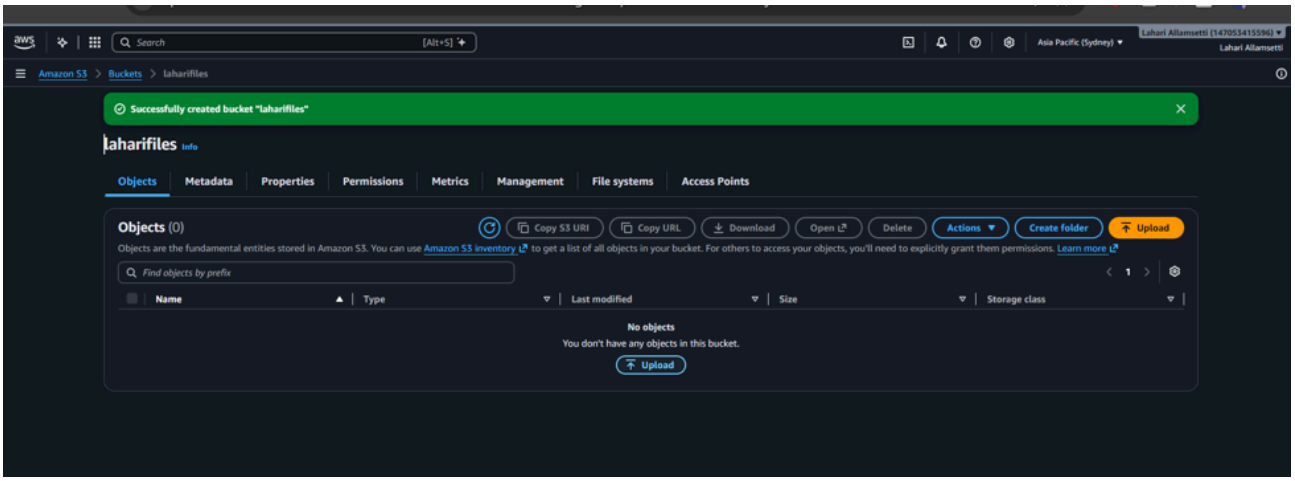
The Amazon S3 page opens with the short description of the service. In the **Create a bucket** box on the right, the orange **Create bucket** button is clicked. The page also shows the pricing information: there is no minimum fee, and charges depend on the usage.

Step 3: General configuration



Under **General configuration** the AWS Region is Asia Pacific (Sydney) ap-southeast-2 and the bucket type is **General purpose**, which suits most uses. The **Global namespace** is selected, and a **Bucket name** is typed in. A bucket name must be 3 to 63 characters long, use only lowercase letters, numbers, periods and hyphens, and be unique across all AWS accounts, which is why the bucket was finally created with the unique name **laharifiles**.

Step 4: Encryption and creating the bucket



Further down, **Default encryption** uses **server-side encryption with Amazon S3 managed keys (SSE-S3)**, so every uploaded object is encrypted automatically, and **Bucket Key** is enabled. The other settings are left at their defaults, and the orange **Create bucket** button at the bottom right is clicked.

Step 5: Bucket created

The green banner confirms **Successfully created bucket "laharifiles"**. The bucket page has the tabs Objects, Metadata, Properties, Permissions, Metrics, Management and Access Points. The bucket is empty (**Objects (0)**), and files and folders can now be added with **Upload** or **Create folder**.

Result: A storage space was created with Amazon S3 by opening the S3 service, choosing **Create bucket**, giving a unique bucket name, keeping the default encryption, and creating the bucket **laharifiles**.

CONCLUSION

In this practical a Linux virtual machine was launched and used from the terminal, a Windows Server instance and a graphical Ubuntu instance were created and reached through Remote Desktop, and an S3 bucket was created for storing data. Together these show how compute (EC2) and storage (S3) services are set up on AWS.